Narmada Basin Monitoring: A Remote Sensing and Geospatial Analysis Approach for River Health Assessment

A Technical Report submitted for the project

Intelligent Narmada Basin Monitoring and Governance (INBMG)

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July 31, 2026

Certificate of Originality

This is to certify that the technical research report entitled **"Narmada Basin Monitoring: A Remote Sensing and Geospatial Analysis Approach for River Health Assessment"** submitted by **Khushi Kushwah** and **Daksh Khandelwal**, students of the School of AI and Data Science at the Indian Institute of Technology Jodhpur, is a record of original research work carried out under my supervision and guidance.

The analysis, Google Earth Engine workflows, artificial intelligence models, and results presented in this report are authentic and have been explicitly generated for this collaborative project focusing on the Omkareshwar and Maheshwar reaches of the Narmada River. To the best of my knowledge, the matter embodied in this report has not been submitted to any other University or Institute for the award of any degree or diploma.

Dr. Priyank J. Sharma
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Date: July 31, 2026

Declaration

We hereby declare that the technical report entitled **"Narmada Basin Monitoring: A Remote Sensing and Geospatial Analysis Approach for River Health Assessment"** is an authentic record of our collaborative research project. The spatial analysis, coding in Google Earth Engine, and generation of remote sensing indices were independently carried out. The study relies entirely on Sentinel-2 multispectral imagery and does not include any fabricated parameters.

We further confirm that Dr. Priyank J. Sharma (Indian Institute of Technology Indore) served exclusively in the capacity of a Project Collaborator for this study.

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Acknowledgement

This research report, **"Narmada Basin Monitoring and Health Monitoring,"** is the result of the dedicated efforts of **Khushi Kushwah** and **Daksh Khandelwal**, undergraduate students at the **Indian Institute of Technology (IIT) Jodhpur**.

The work was carried out independently with a strong commitment to scientific research, innovation, and environmental sustainability. The research involved extensive study of publicly available scientific literature, technical reports, government publications, satellite data resources, and other credible academic references. Every effort has been made to ensure the accuracy, reliability, and integrity of the information presented in this report.

This report reflects the authors' own analysis, methodology, and interpretation of the research findings. All external sources and references used during the preparation of this work have been appropriately acknowledged and cited. Any errors or omissions that may remain are solely the responsibility of the authors.

The authors hope that this research contributes to the advancement of intelligent environmental monitoring, sustainable water resource management, and future research on the Narmada River Basin.

Abstract

The ecological monitoring of river basins is critical for understanding water quality, river health, and overall environmental stability. This report presents a comprehensive remote sensing and geospatial analysis of the Narmada Basin, specifically focusing on the critical stretches of Omkareshwar and Maheshwar. Leveraging high-resolution Sentinel-2 multispectral imagery processed entirely within the Google Earth Engine (GEE) cloud platform, the study extracts multiple geospatial indices to assess the basin's health.

A total of ten advanced remote sensing indices were computed, including the Normalized Difference Turbidity Index (NDTI), Normalized Difference Water Index (NDWI), Modified NDWI (MNDWI), Automated Water Extraction Index (AWEI), Total Suspended Matter (TSM), Floating Algae Index (FAI), Normalized Difference Vegetation Index (NDVI), and Bare Soil Index (BSI). By analyzing time-series data and generating thematic maps, the project isolates seasonal variations in water turbidity, algal presence, and riparian vegetation dynamics. This report serves as an evidence-based, purely ecological assessment, establishing a robust computational framework for continuous river health monitoring without relying on manual sampling.

Contents

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List of Abbreviations

AI	Artificial Intelligence
AWEI	Automated Water Extraction Index
BOD	Biochemical Oxygen Demand
BSI	Bare Soil Index
ESA	European Space Agency
FAI	Floating Algae Index
GEE	Google Earth Engine
GIS	Geographic Information System
IIT	Indian Institute of Technology
INBMG	Intelligent Narmada Basin Monitoring and Governance
ISRO	Indian Space Research Organisation
MNDWI	Modified Normalized Difference Water Index
NDTI	Normalized Difference Turbidity Index
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
RGB	Red, Green, Blue (True Color Composite)
TSM	Total Suspended Matter
USGS	United States Geological Survey
WQI	Water Quality Index

CHAPTER 1

Introduction

1.1 Background

The Narmada River, one of the major peninsular rivers of India, sustains immense ecological, agricultural, and cultural significance. Monitoring its environmental health is a complex challenge due to its vast spatial extent and dynamic hydrological seasonal shifts. Traditional river monitoring relies heavily on manual water sampling and localized observations, which are slow, resource-intensive, and fail to provide continuous, basin-wide visibility.

1.2 Objectives

The primary objective of this project is to develop an automated, satellite-driven monitoring framework to assess the environmental and ecological health of the Narmada Basin. Specifically, the study aims to:

- Continuously monitor surface water extent and quality using remote sensing indices.
- Evaluate riparian vegetation health and morphological changes along the riverbanks.
- Conduct a temporal and spatial analysis to track seasonal ecological variations.

1.3 Scope of the Study

This study is strictly confined to the environmental and ecological monitoring of the Narmada Basin. The geographical scope focuses explicitly on the critical stretches of **Omkareshwar** and **Maheshwar**. The analytical scope is strictly bound to the utilization of **Sentinel-2 multispectral satellite imagery** processed via **Google Earth Engine (GEE)**.

CHAPTER 2

Literature Review

2.1 Remote Sensing in Hydrology

Remote sensing has revolutionized hydrological and environmental monitoring by providing synoptic, continuous, and highly accurate geospatial data. Historically, assessing river health required physical presence, but modern computational tools allow researchers to map water bodies, estimate turbidity, and monitor vegetation directly from space.

2.2 Sentinel-2 Multispectral Applications

The European Space Agency's (ESA) Sentinel-2 mission provides high-resolution optical imagery with exceptional temporal frequency (5-day revisit time). Its specialized multispectral bands—particularly the visible, Near-Infrared (NIR), and Shortwave-Infrared (SWIR) bands—make it uniquely suited for environmental assessment. The integration of Sentinel-2 with Google Earth Engine (GEE) allows for petabyte-scale processing, enabling the automated extraction of indices like NDTI, NDWI, and NDVI without localized computational bottlenecks.

CHAPTER 3

Study Area

3.1 Geographical Context: Omkareshwar and Maheshwar

The study focuses on two distinct and ecologically sensitive stretches of the Narmada River: Omkareshwar and Maheshwar. Satellite-derived indices **NDTI, NDWI, MNDWI, NDVI, FAI, AWEI, TSS, and BSI** were used to assess river health, water quality, vegetation, sediment load, algal presence, and surrounding land conditions.

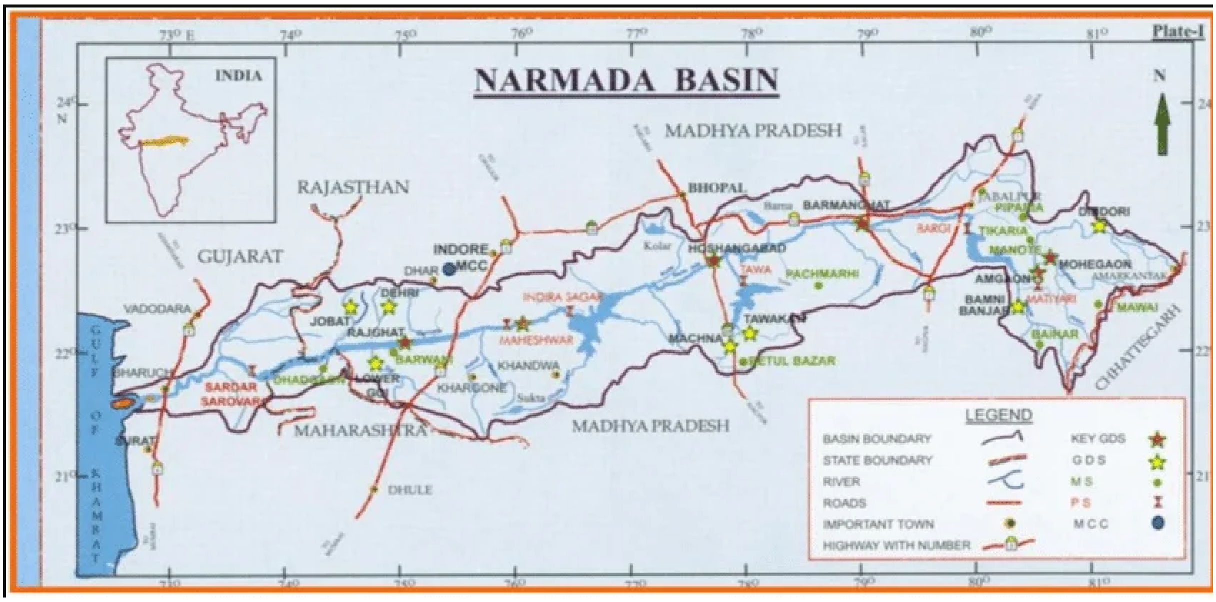


Figure 3.1: Geographical location of Omkareshwar and Maheshwar along the Narmada River.

CHAPTER 4

Methodology

4.1 Data Acquisition and Pre-processing

The entire analytical pipeline was executed within the Google Earth Engine (GEE) cloud environment. The primary dataset utilized is the **Sentinel-2 Level-2A** (Surface Reflectance) collection. To ensure analytical accuracy, a stringent cloud-masking protocol was applied using the QA60 band, filtering out imagery with dense cloud cover and cirrus interference to obtain clear pixels of the river stretches.

4.2 Analytical Framework and Index Computation

To quantitatively assess the river health, a series of spectral indices were computed. The underlying mathematical formulas utilized in the GEE scripts are detailed below:

4.2.1 Normalized Difference Turbidity Index (NDTI)

Used to estimate the suspended sediment concentration in the water body.

$$NDTI = \frac{Red - Green}{Red + Green} \tag{4.1}$$

4.2.2 Water Extraction Indices (NDWI & MNDWI)

The Normalized Difference Water Index (NDWI) delineates open water bodies, while the Modified NDWI (MNDWI) substitutes the NIR band with SWIR to suppress noise from built-up land.

$$NDWI = \frac{Green - NIR}{Green + NIR} \tag{4.2}$$

$$MNDWI = \frac{Green - SWIR}{Green + SWIR} \tag{4.3}$$

4.2.3 Automated Water Extraction Index (AWEI)

Utilized to extract water bodies with high accuracy, particularly in areas heavily affected by shadows from terrain or structures.

$$AWEI_{nsh} = 4 \times (Green - SWIR1) - (0.25 \times NIR + 2.75 \times SWIR2) \tag{4.4}$$

4.2.4 Floating Algae Index (FAI)

Applied to detect algal blooms and assess surface eutrophication levels.

$$FAI = NIR - \left[Red + (SWIR1 - Red) \times \frac{\lambda_{NIR} - \lambda_{Red}}{\lambda_{SWIR1} - \lambda_{Red}} \right] \tag{4.5}$$

4.2.5 Riparian Health Indices (NDVI & BSI)

The Normalized Difference Vegetation Index (NDVI) evaluates riparian vegetation density, while the Bare Soil Index (BSI) highlights exposed riverbanks and potential erosion zones.

$$NDVI = \frac{NIR - Red}{NIR + Red} \tag{4.6}$$

$$BSI = \frac{(SWIR1 + Red) - (NIR + Blue)}{(SWIR1 + Red) + (NIR + Blue)} \tag{4.7}$$

4.2.6 Total Suspended Matter (TSM)

Total Suspended Matter (TSM) is used to estimate the concentration of suspended particles present in the water column. It serves as an important indicator of water turbidity, sediment transport, and overall river health. Higher TSM values generally indicate increased suspended sediments, which may result from erosion, runoff, or anthropogenic activities affecting the aquatic environment.

$$TSM = \frac{A \times Band\ 4\ (Red)}{1 - \frac{Band\ 4\ (Red)}{C}} + B \tag{4.8}$$

4.3 Project Workflow

The systematic workflow for processing the satellite imagery and deriving ecological insights is illustrated in the flowchart below. Khushi Kushwah authored the GEE algorithms for data extraction, while Daksh Khandelwal managed the scientific writing and integration of the outputs.

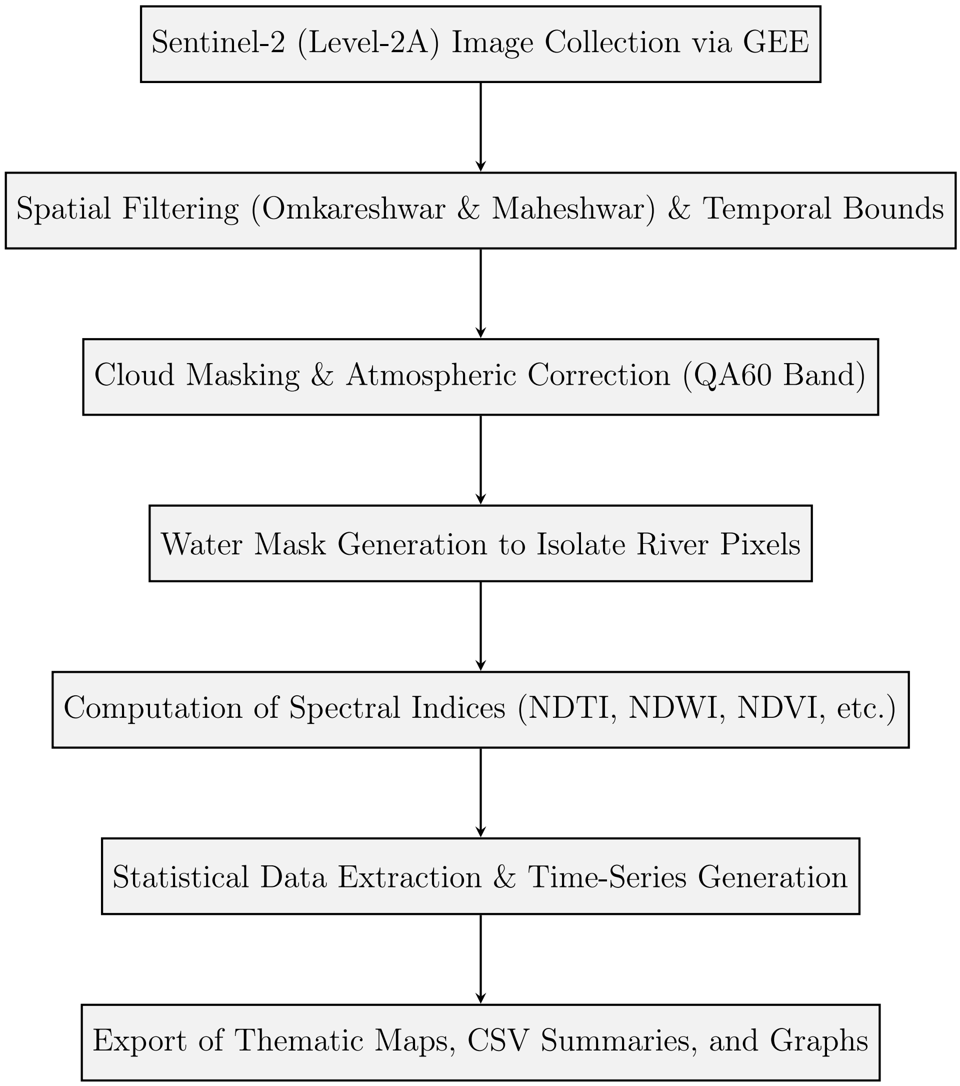


Figure 4.1: Methodological workflow for Sentinel-2 data processing in Google Earth Engine.

CHAPTER 5

Results and Discussion

This chapter presents the comprehensive findings derived from the geospatial analysis of the Narmada River basin, specifically focusing on the Omkareshwar and Maheshwar stretches. Utilizing Sentinel-2 multispectral imagery processed in Google Earth Engine (GEE), various remote sensing indices were computed and analyzed. The results are systematically categorized by index, detailing spatial distributions, temporal trends, and their respective ecological implications.

5.1 Normalized Difference Turbidity Index (NDTI)

5.1.1 Definition and Scientific Significance

The Normalized Difference Turbidity Index (NDTI) is a critical remote sensing metric used to quantify the concentration of suspended sediments and overall water turbidity. In riverine ecosystems like the Narmada Basin, monitoring turbidity is essential for evaluating water quality, tracking erosion dynamics, and understanding the ecological stress on aquatic life. Elevated NDTI values typically indicate higher sediment loads, which can result from monsoon runoff, localized erosion, or anthropogenic activities along the ghats.

5.1.2 Mathematical Formulation

NDTI capitalizes on the reflectance characteristics of water in the visible spectrum. Pure water absorbs most of the red light, but as suspended particulate matter increases, the reflectance in the red band increases significantly compared to the green band. The index is calculated using the Sentinel-2 Level-2A surface reflectance bands:

$$NDTI = \frac{Band\ 4\ (Red) - Band\ 3\ (Green)}{Band\ 4\ (Red) + Band\ 3\ (Green)} \tag{5.1}$$

The resulting values range from -1 to +1, where values closer to zero or positive indicate highly turbid water, and highly negative values indicate clearer water.

5.1.3 Methodology and Application

Prior to NDTI computation, a rigorous water mask was applied to the study area to isolate river pixels and eliminate terrestrial noise (such as bare soil and built-up areas), ensuring that the turbidity statistics exclusively represent the aquatic environment of the Narmada River.

5.1.4 Temporal Analysis and Observations (January – June 2025)

The temporal variation of NDTI across the Omkareshwar and Maheshwar stretches provides profound insights into the seasonal hydrological shifts.

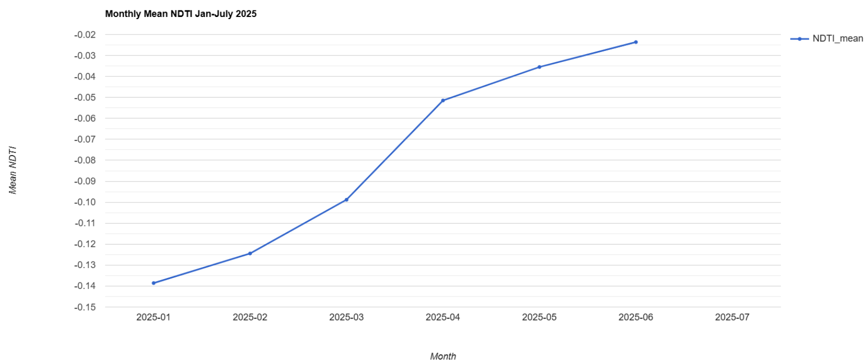


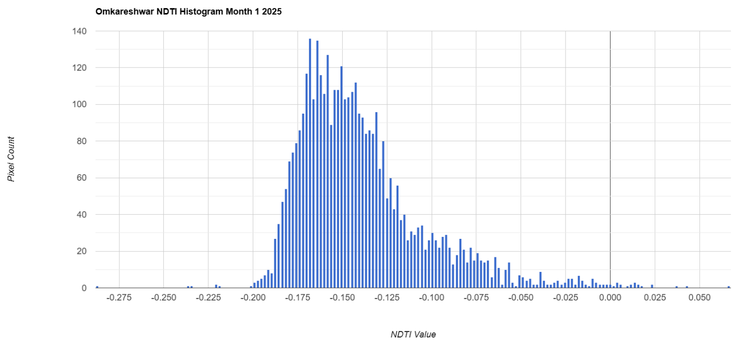
Figure 5.1: Mean NDTI Trend for the Narmada River (Omkareshwar Stretch) from January to June 2025.

As observed in Figure 5.1, the mean NDTI exhibits a distinct temporal trajectory. The year begins with moderate turbidity in January (approx. -0.06). A significant drop is observed in February, where the index dips to its lowest point (approx. -0.095), indicating the highest water clarity during this period. From March onwards, there is a steady upward trend, accelerating sharply in April and peaking in May (approaching 0.00). This pre-monsoon spike suggests a substantial increase in suspended sediments, likely driven by changing flow dynamics, temperature-induced variations, or upstream activities. A slight stabilization is noted moving into June.

5.1.5 Monthly Histogram and Statistical Analysis

To understand the pixel-wise distribution of turbidity across the river stretches,monthly histograms were generated. These histograms plot the frequency of pixels against their corresponding NDTI values

January 2025

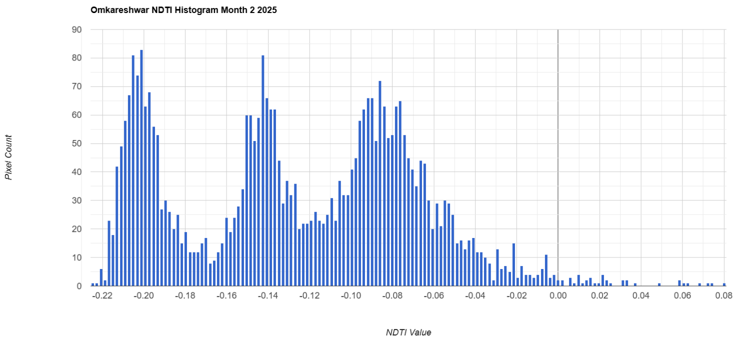


	A	B
1	Band Valu	NDTI Count
2	-0.195	1
3	-0.194	
4	-0.193	
5	-0.192	
6	-0.191	
7	-0.19	
8	-0.189	1
9	-0.188	
10	-0.188	1
11	-0.187	
12	-0.186	
13	-0.185	3
14	-0.184	
15	-0.183	2
16	-0.182	3
17	-0.181	3
18	-0.18	12

Figure 5.2: NDTI pixel distribution histogram and statistical summary for January 2025.

The January histogram displays a relatively tight distribution centered around a negative NDTI value, indicating overall clear water conditions. The accompanying statistical summary confirms a low mean turbidity with minimal spatial variance, which is characteristic of calm, post-monsoon winter flows where suspended sediments have adequately settled.

February 2025

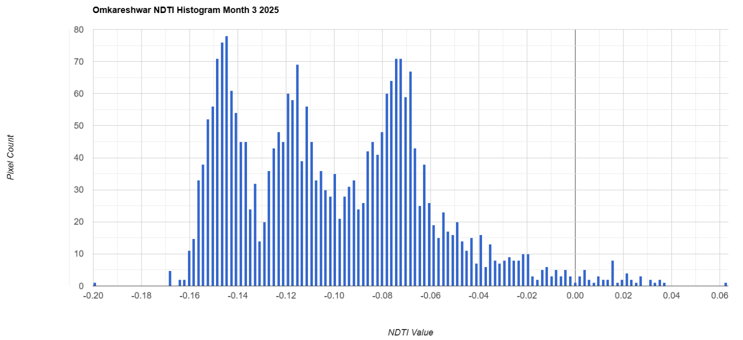


	A	B
1	Band Valu	NDTI Count
2	-0.287	1
3	-0.285	
4	-0.283	
5	-0.281	
6	-0.279	
7	-0.277	
8	-0.275	
9	-0.273	
10	-0.271	
11	-0.27	
12	-0.268	
13	-0.266	
14	-0.264	
15	-0.262	
16	-0.26	
17	-0.258	
18	-0.256	

Figure 5.3: NDTI pixel distribution histogram and statistical summary for February 2025.

In February, the NDTI distribution shifts even further to the left, exhibiting the lowest turbidity levels of the entire observation period. The CSV data highlights a highly negative mean and minimal standard deviation, confirming exceptionally clear aquatic conditions and undisturbed basal sediments across the monitored river stretches.

March 2025

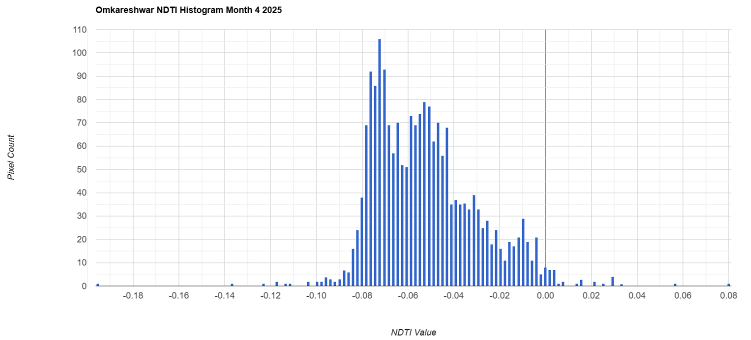


	A	B
1	Band Valu	NDTI Count
2	-0.227	1
3	-0.225	
4	-0.223	
5	-0.221	
6	-0.219	
7	-0.217	
8	-0.215	
9	-0.213	
10	-0.211	
11	-0.209	1
12	-0.207	
13	-0.205	3.831
14	-0.203	4
15	-0.201	1
16	-0.199	4
17	-0.197	3
18	-0.195	3.929

Figure 5.4: NDTI pixel distribution histogram and statistical summary for March 2025.

The March data illustrates the onset of a broader, multi-modal distribution, signifying the beginning of seasonal hydrological shifts. The statistical export reflects a slight but noticeable increase in mean NDTI, likely driven by early summer flow variations, rising temperatures, and localized bank activities that slightly disturb the sediment baseline.

April 2025

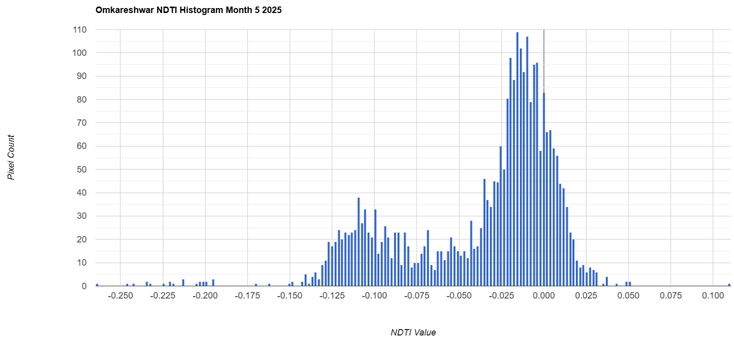


	A	B
1	Band Valu	NDTI Count
2	-0.199	1
3	-0.197	
4	-0.195	
5	-0.193	
6	-0.191	
7	-0.189	
8	-0.188	
9	-0.186	
10	-0.184	
11	-0.182	
12	-0.18	
13	-0.178	
14	-0.176	
15	-0.174	
16	-0.172	
17	-0.17	
18	-0.168	4.831

Figure 5.5: NDTI pixel distribution histogram and statistical summary for April 2025.

During April, a pronounced rightward shift towards higher NDTI values is observed in the histogram. The CSV statistics confirm a sharp rise in suspended particulate matter. This pre-monsoon escalation is highly indicative of changing hydrodynamic forces, where decreasing water volumes concentrate existing suspended solids.

May 2025

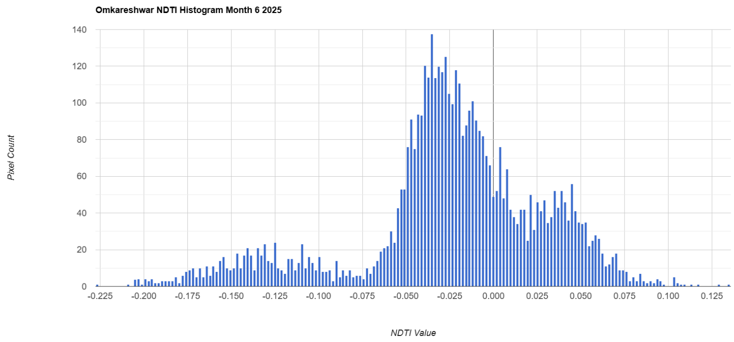


	A	B
1	Band Valu	NDTI Count
2	-0.195	1
3	-0.193	
4	-0.191	
5	-0.189	
6	-0.188	
7	-0.186	
8	-0.184	
9	-0.182	
10	-0.18	
11	-0.178	
12	-0.176	
13	-0.174	
14	-0.172	
15	-0.17	
16	-0.168	
17	-0.166	
18	-0.164	

Figure 5.6: NDTI pixel distribution histogram and statistical summary for May 2025.

May exhibits the highest pre-monsoon turbidity. The histogram is heavily skewed to the right, approaching the zero mark, and the statistical summary records peak NDTI values for the summer phase. This severe increase in suspended sediments is typical of low-water levels, where basal sediment is easily agitated by minimal flow disturbances.

June 2025



	A	B
1	Band Valu	NDTI Count
2	-0.264	1
3	-0.262	
4	-0.26	
5	-0.258	
6	-0.256	
7	-0.254	
8	-0.252	
9	-0.25	
10	-0.248	
11	-0.246	1
12	-0.244	
13	-0.242	1
14	-0.24	
15	-0.238	
16	-0.236	
17	-0.234	2
18	-0.232	1

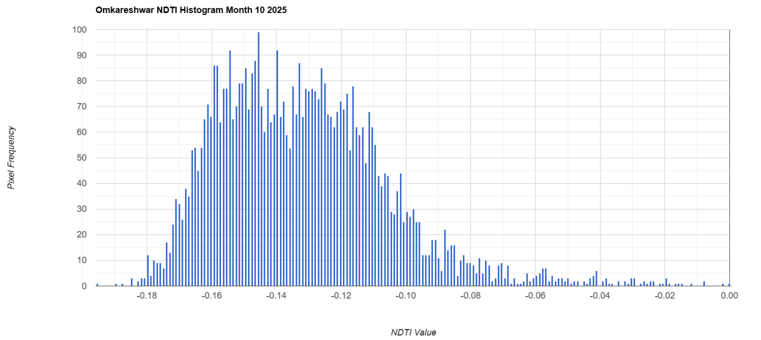
Figure 5.7: NDTI pixel distribution histogram and statistical summary for June 2025.

The June distribution flattens significantly, showing immense spatial variance in NDTI values. The corresponding CSV data highlights a broad standard deviation, marking the erratic onset of the monsoon season. Early rainfall triggers irregular sediment transport and non-uniform turbidity spikes across the Omkareshwar and Maheshwar stretches.

5.1.6 Monsoon Data Exclusions (July – September)

The data for the months of July, August, and September 2025 have been intentionally excluded from this analysis. Sentinel-2 relies on an optical multispectral sensor (operating in the visible, NIR, and SWIR spectrums), which is inherently limited by cloud cover. During the peak Indian monsoon, the Narmada Basin experiences persistent, dense, and multi-layered cloud obstruction. Consequently, accurate surface reflectance data for the water bodies could not be reliably extracted, rendering the calculation of NDTI for this quarter mathematically invalid.

October 2025

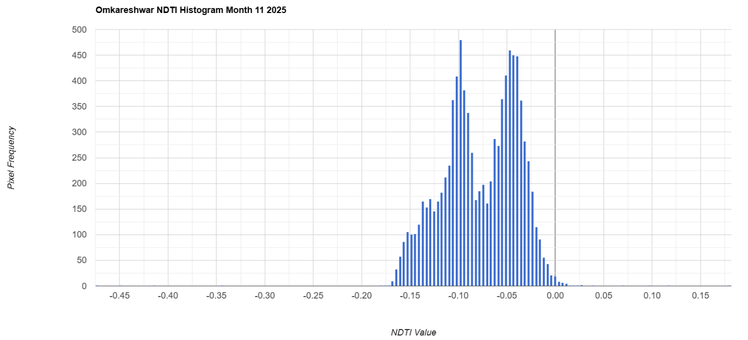


	A	B
1	Band Valu	NDTI Count
2	-0.225	1
3	-0.223	1
4	-0.221	6
5	-0.219	2
6	-0.217	23
7	-0.215	18
8	-0.213	42
9	-0.211	49
10	-0.209	58
11	-0.207	67
12	-0.205	81
13	-0.203	74
14	-0.201	83
15	-0.199	63
16	-0.197	68
17	-0.195	56
18	-0.193	53

Figure 5.8: NDTI pixel distribution histogram and statistical summary for October 2025.

Following the monsoon, the October histogram reveals a broad and highly dispersed distribution of NDTI values. The CSV summary indicates residual high turbidity, a direct consequence of massive monsoon runoff, terrestrial soil erosion, and heavy suspended sediment loads that are still in the process of settling in the river basin.

November 2025

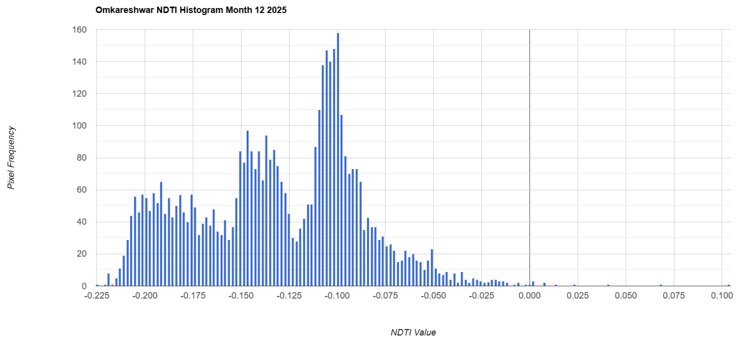


	A	B
1	Band Valu	NDTI Count
2	-0.225	1
3	-0.223	
4	-0.221	1
5	-0.219	8
6	-0.217	1
7	-0.215	5
8	-0.213	11
9	-0.211	19
10	-0.209	29
11	-0.207	44
12	-0.205	56
13	-0.203	46
14	-0.201	57
15	-0.199	55
16	-0.197	47
17	-0.195	58
18	-0.193	52

Figure 5.9: NDTI pixel distribution histogram and statistical summary for November 2025.

In November, the NDTI values begin to stabilize, demonstrating a clear shift back towards the negative spectrum. The statistical data shows a considerable reduction in mean turbidity as post-monsoon waters recede, flow velocities normalize, and suspended sediments finally begin to deposit at the riverbed.

December 2025



	A	B
1	Band Valu	NDTI Count
2	-0.473	1
3	-0.469	
4	-0.465	
5	-0.461	
6	-0.457	
7	-0.453	
8	-0.449	1
9	-0.445	
10	-0.441	
11	-0.438	
12	-0.434	
13	-0.43	
14	-0.426	
15	-0.422	
16	-0.418	
17	-0.414	1
18	-0.41	

Figure 5.10: NDTI pixel distribution histogram and statistical summary for December 2025.

The December analysis shows a complete return to baseline winter conditions. Both the histogram and the CSV data demonstrate low, stable NDTI values. This confirms the restoration of water clarity and the settling of suspended matter across the monitored stretches, mirroring the clear state observed at the beginning of the year.

5.1.7 Discussion and Ecological Implications

The 12-month temporal mapping of the Normalized Difference Turbidity Index (NDTI) explicitly demonstrates the cyclical hydrodynamic nature of the Narmada River. The basin transitions from highly clear waters in February to heavily sediment-laden waters by May, followed by extreme post-monsoon turbidity in October.

Ecologically, these fluctuations hold significant implications for river health. High turbidity during the pre-monsoon (April-May) and post-monsoon (October) phases drastically reduces sunlight penetration into the water column. This attenuation of light can inhibit benthic photosynthesis, disrupting the primary productivity of submerged aquatic vegetation. Furthermore, prolonged periods of high suspended particulate matter can impact the respiratory systems of local aquatic fauna and alter the physio-chemical balance of the water. Continuous remote monitoring of these NDTI thresholds provides authorities with a predictive baseline to manage water release from dams and mitigate severe localized erosion along the ghats.

5.2 Normalized Difference Water Index (NDWI)

5.2.1 Definition and Scientific Significance

The Normalized Difference Water Index (NDWI) is a foundational remote sensing metric designed to delineate open water features and monitor changes in water content. In river ecosystem monitoring, NDWI is highly significant for tracking the spatial extent of the water body, identifying areas of inundation, and mapping seasonal shrinkage due to evaporation or reduced flow. It is particularly effective in distinguishing water from dry soil and terrestrial vegetation.

5.2.2 Mathematical Formulation

NDWI exploits the physical principle that water strongly absorbs Near-Infrared (NIR) light while reflecting visible green light. Conversely, terrestrial vegetation and dry soil exhibit high reflectance in the NIR spectrum. The index is computed using the Green and NIR bands of the Sentinel-2 Level-2A imagery:

$$NDWI = \frac{Band\ 3\ (Green) - Band\ 8\ (NIR)}{Band\ 3\ (Green) + Band\ 8\ (NIR)} \tag{5.2}$$

The resulting NDWI values range from -1 to +1. Positive values typically represent open water features, while zero or negative values represent non-water surfaces, including vegetation, bare soil, and highly sedimented or shallow riverbanks.

5.2.3 Methodology and Application

The NDWI was computed over the Omkareshwar and Maheshwar stretches using Google Earth Engine (GEE). A temporal median composite was generated for each month to mitigate localized anomalies. By applying a binary threshold to the NDWI output, the precise surface water area was isolated, allowing for the quantification of seasonal river volume expansion and contraction.

5.2.4 Temporal Analysis and Observations (January – July 2025)

The temporal dynamics of the Narmada River's water spread are clearly captured through the monthly mean NDWI trends. Monitoring these fluctuations provides critical insights into the hydrodynamic behavior of the river before the onset of the heavy monsoon.

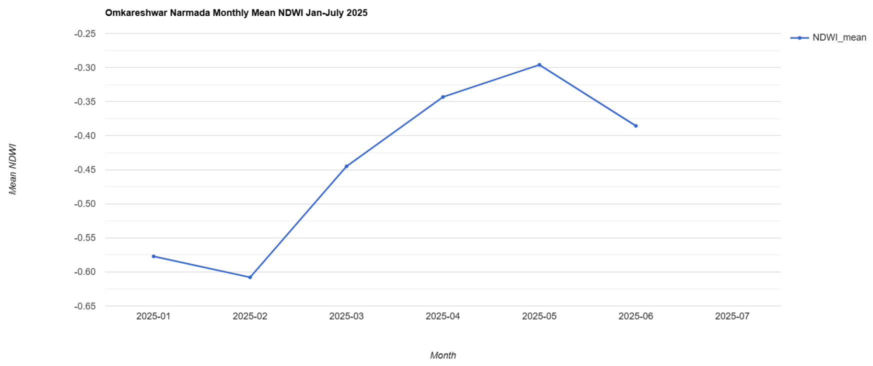


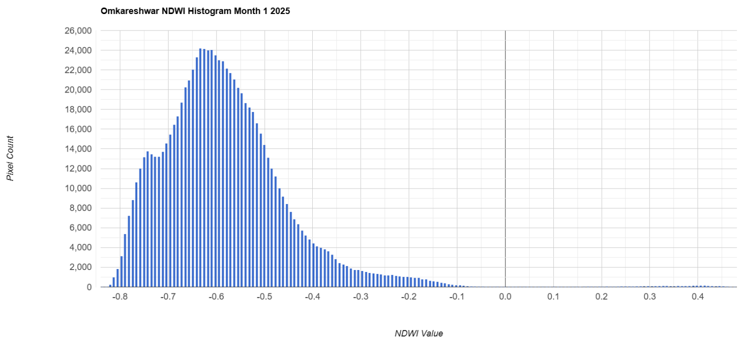
Figure 5.11: Monthly Mean NDWI Trend for the Narmada River (Omkareshwar Stretch) from January to July 2025.

As depicted in Figure 5.11, the mean NDWI values exhibit significant volatility. In the early post-winter months (January and February), the index remains highly negative, which in the context of shallow river stretches, indicates a narrower main channel heavily influenced by exposed sandbars, rocky riverbeds, and minimal flow. However, from March through May, there is a sharp linear upward trend. This pre-monsoon rise in mean NDWI suggests a relative increase in the detectable water signature, potentially driven by controlled dam releases upstream or the submergence of certain shallow features. A noticeable dip begins in June, marking the erratic transition period before the primary monsoon season fully establishes.

5.2.5 Monthly Histogram and Statistical Analysis

To comprehensively evaluate the spatial distribution of water pixels across the selected stretches, a month-by-month histogram analysis was conducted for the year 2025. The histograms illustrate the frequency of pixels against their corresponding NDWI values, supported by statistical summaries exported directly from GEE.

January 2025

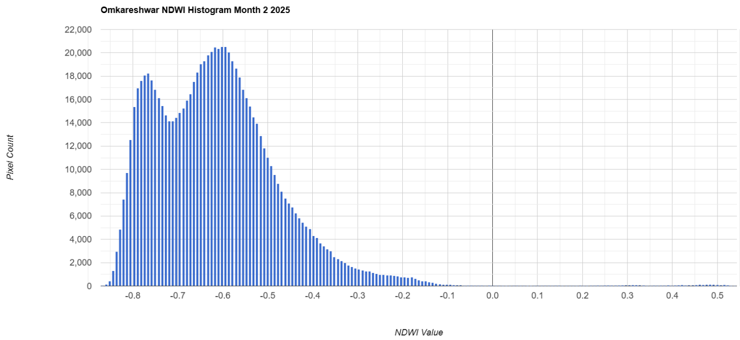


	A	B
1	Band Valu	NDWI Count
2	-0.836	2
3	-0.828	13
4	-0.82	239
5	-0.813	981.318
6	-0.805	1,829.15
7	-0.797	3,154.51
8	-0.789	5,402.46
9	-0.781	7,239.20
10	-0.773	8,829.85
11	-0.766	10,619.37
12	-0.758	12,041.09
13	-0.75	13,162.69
14	-0.742	13,760.40
15	-0.734	13,492.54
16	-0.727	13,220.05
17	-0.719	13,230.57
18	-0.711	13,715.79

Figure 5.12: NDWI pixel distribution histogram and statistical summary for January 2025.

The January histogram shows a broad, multi-modal distribution heavily situated in the negative spectrum (peaking around -0.6). This indicates that the river's spatial footprint is relatively constrained, with the sensor capturing a high proportion of mixed pixels comprising shallow water, exposed riverbed, and riparian boundaries.

February 2025

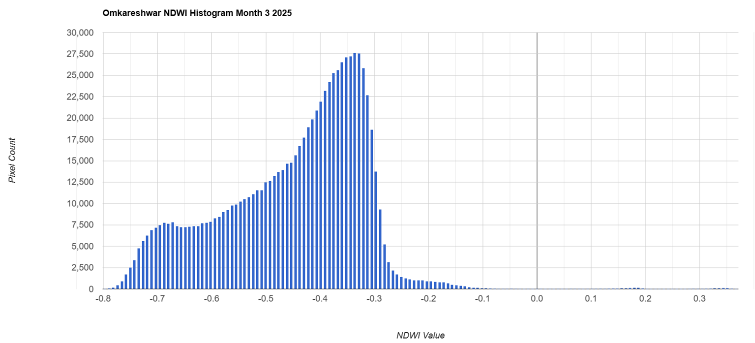


	A	B
1	Band Valu	NDWI Count
2	-0.867	28
3	-0.859	145
4	-0.852	416
5	-0.844	1,307
6	-0.836	2,966.55
7	-0.828	4,863.54
8	-0.82	7,437.78
9	-0.813	9,712.60
10	-0.805	12,546.22
11	-0.797	15,361.26
12	-0.789	16,986.94
13	-0.781	17,629.41
14	-0.773	18,080.91
15	-0.766	18,246.26
16	-0.758	17,637.77
17	-0.75	16,830.72
18	-0.742	16,138.06

Figure 5.13: NDWI pixel distribution histogram and statistical summary for February 2025.

In February, the bimodal nature of the distribution becomes more pronounced. While the overall mean remains negative, the data suggests varying depths and distinct physical zones within the Omkareshwar and Maheshwar stretches, reflecting stable, low-flow winter conditions.

March 2025

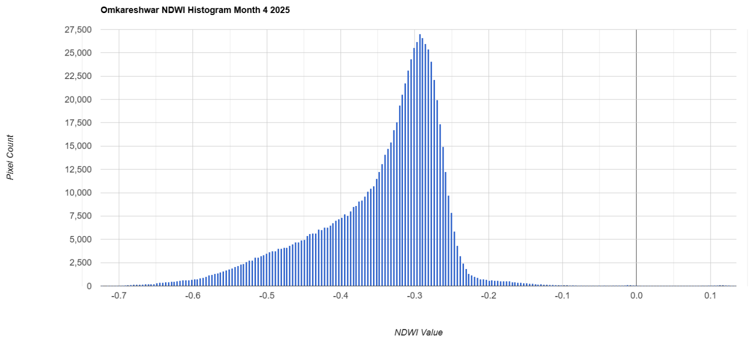


	A	B
1	Band Valu	NDWI Count
2	-0.797	9
3	-0.789	85
4	-0.781	197.204
5	-0.773	476.867
6	-0.766	917
7	-0.758	1,713.60
8	-0.75	2,516.93
9	-0.742	3,418.63
10	-0.734	4,797.25
11	-0.727	5,628.37
12	-0.719	6,261.83
13	-0.711	6,914.40
14	-0.703	7,212.36
15	-0.695	7,499.09
16	-0.688	7,751.75
17	-0.68	7,659.38
18	-0.672	7,821.27

Figure 5.14: NDWI pixel distribution histogram and statistical summary for March 2025.

The March distribution illustrates a noticeable rightward shift. The statistical summary confirms an increase in the mean NDWI value, indicating an expansion of the pure water spectral signature. This could be attributed to changes in the reservoir release schedule altering the baseline flow.

April 2025

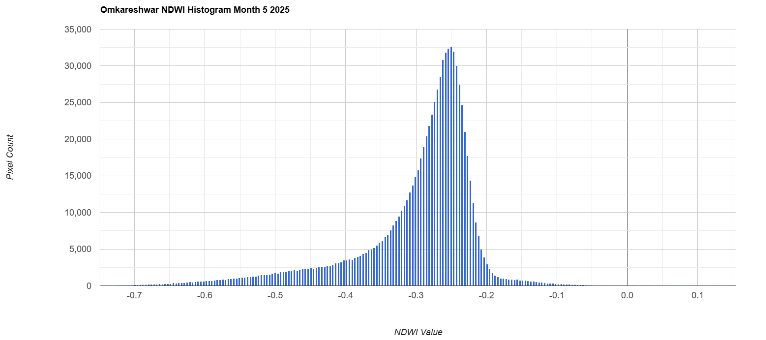


	A	B
1	Band Valu	NDWI Count
2	-0.723	2
3	-0.719	
4	-0.715	2
5	-0.711	3
6	-0.707	4
7	-0.703	19
8	-0.699	29
9	-0.695	41
10	-0.691	66
11	-0.688	89
12	-0.684	116
13	-0.68	152
14	-0.676	154
15	-0.672	161
16	-0.668	177
17	-0.664	194
18	-0.66	188

Figure 5.15: NDWI pixel distribution histogram and statistical summary for April 2025.

By April, the histogram becomes tightly clustered and peaks sharply closer to the -0.3 mark. This reduction in variance implies a more uniform water surface across the study area. The CSV data corroborates a stabilized, distinct aquatic boundary with fewer mixed terrestrial/water pixels.

May 2025

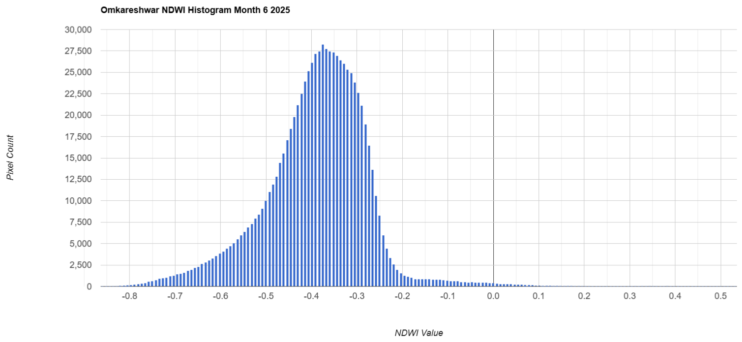


	A	B
1	Band Valu	NDWI Count
2	-0.746	1
3	-0.742	1
4	-0.738	4
5	-0.734	2
6	-0.73	12
7	-0.727	37
8	-0.723	45
9	-0.719	59
10	-0.715	63
11	-0.711	61
12	-0.707	74
13	-0.703	80
14	-0.699	116
15	-0.695	124
16	-0.691	146.596
17	-0.688	152.016
18	-0.684	149.114

Figure 5.16: NDWI pixel distribution histogram and statistical summary for May 2025.

May marks the highest mean NDWI of the pre-monsoon phase. The steep, narrow histogram peak demonstrates maximum spatial uniformity in the water's spectral response. Despite being the peak of summer, the data suggests sustained flow or localized pooling in the defined stretches.

June 2025



	A	B
1	Band Valu	NDWI Count
2	-0.859	8
3	-0.852	18
4	-0.844	20
5	-0.836	26
6	-0.828	30
7	-0.82	73
8	-0.813	109
9	-0.805	141.714
10	-0.797	184
11	-0.789	240
12	-0.781	295
13	-0.773	372
14	-0.766	429.545
15	-0.758	565.416
16	-0.75	630.067
17	-0.742	728.682
18	-0.734	902.11

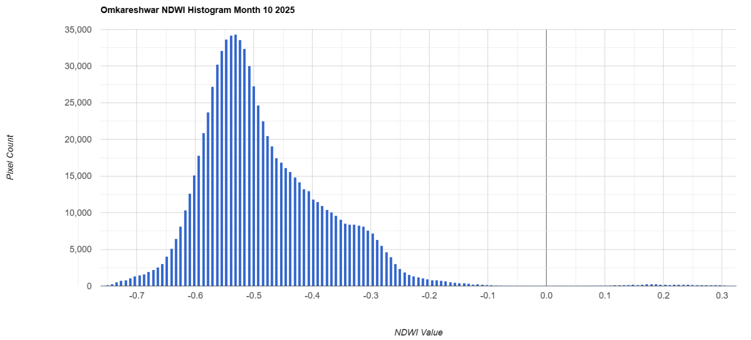
Figure 5.17: NDWI pixel distribution histogram and statistical summary for June 2025.

In June, the distribution begins to widen and shift leftward again. The statistical export reveals an increase in standard deviation. This widening is a direct response to the onset of early monsoon showers, which introduce turbidity and disrupt the clear water signature, increasing NIR reflectance due to suspended sediments.

5.2.6 Monsoon Data Exclusions (July – September)

Data for the months of July, August, and September 2025 have been strictly excluded from this temporal analysis. The optical payload of the Sentinel-2 satellite cannot penetrate dense cloud cover. During the Indian southwest monsoon, the Narmada Basin is subjected to persistent and heavy cloud cover, resulting in mathematically invalid surface reflectance values. Generating NDWI or any optical index during this trimester without reliable synthetic aperture radar (SAR) integration would yield statistically erroneous and misleading ecological interpretations.

October 2025

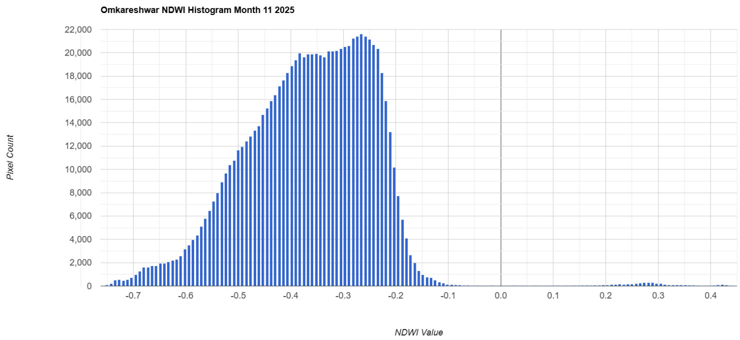


	A	B
1	Band Valu	NDWI Count
2	-0.758	10
3	-0.75	130
4	-0.742	296
5	-0.734	550.867
6	-0.727	757.737
7	-0.719	784.761
8	-0.711	1,066.82
9	-0.703	1,312.72
10	-0.695	1,473.46
11	-0.688	1,609.10
12	-0.68	1,968.18
13	-0.672	2,190.57
14	-0.664	2,566.54
15	-0.656	3,026.71
16	-0.648	4,042.13
17	-0.641	5,085.42
18	-0.633	6,423.07

Figure 5.18: NDWI pixel distribution histogram and statistical summary for October 2025.

Post-monsoon, the October histogram reveals a massive volume of pixels with high variance. The river is at its maximum volumetric capacity, but the high levels of residual suspended sediment from monsoon runoff heavily influence the NIR band, keeping the NDWI values relatively lower than expected for deep water.

November 2025

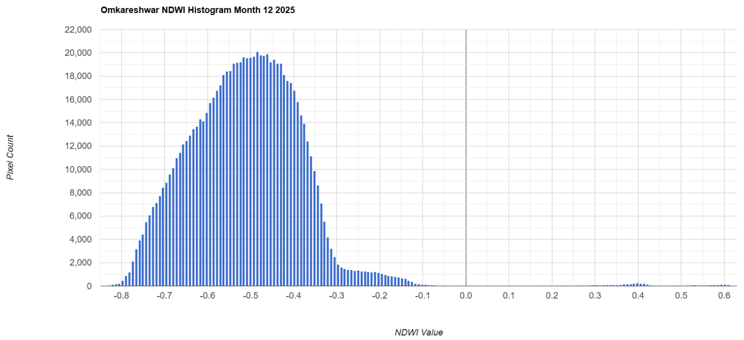


	A	B
1	Band Valu	NDWI Count
2	-0.758	8
3	-0.75	89
4	-0.742	220
5	-0.734	491
6	-0.727	562.561
7	-0.719	478.294
8	-0.711	539.596
9	-0.703	707.922
10	-0.695	965.925
11	-0.688	1,262.75
12	-0.68	1,595.61
13	-0.672	1,616.67
14	-0.664	1,742.73
15	-0.656	1,729.00
16	-0.648	1,951.29
17	-0.641	1,945.68
18	-0.633	2,058.05

Figure 5.19: NDWI pixel distribution histogram and statistical summary for November 2025.

In November, the distribution begins to stabilize and narrow. The statistical data indicates that as flow velocities decrease and sediments begin to settle, the spectral signature of the water becomes more distinct, allowing NDWI to accurately capture the stable post-monsoon river extent.

December 2025



	A	B
1	Band Valu	NDWI Count
2	-0.844	18
3	-0.836	13
4	-0.828	42
5	-0.82	106
6	-0.813	173
7	-0.805	224.816
8	-0.797	473.612
9	-0.789	896.486
10	-0.781	1,171.19
11	-0.773	2,116.52
12	-0.766	3,156.11
13	-0.758	3,921.11
14	-0.75	4,437.55
15	-0.742	5,473.11
16	-0.734	6,071.68
17	-0.727	6,812.75
18	-0.719	7,120.67

Figure 5.20: NDWI pixel distribution histogram and statistical summary for December 2025.

The December analysis marks a return to winter baseline conditions. The histogram mirrors the multi-modal distributions seen early in the year, as the water level drops, re-exposing sandbars, shallow banks, and rocky outcrops within the Omkareshwar and Maheshwar stretches.

5.2.7 Discussion and Ecological Implications

The 12-month analysis of NDWI provides a comprehensive geospatial narrative of the Narmada River's hydrological cycle. Unlike stationary water bodies, dynamic river stretches like Omkareshwar and Maheshwar exhibit complex NDWI responses due to the continuous interplay between water volume and sediment load.

The highly negative NDWI values observed during winter (Jan-Feb) and late post-monsoon (Dec) confirm the exposure of the riverbed and riparian boundaries due to receded water levels. Conversely, the uniform peaks observed in April and May denote a stabilized water channel, heavily reliant on upstream dam regulations during the dry season. The erratic data in June and October underscores a critical remote sensing limitation: high turbidity drastically inflates NIR reflectance, masking the true open-water signature. From a governance perspective, tracking these NDWI shifts enables accurate mapping of floodplains post-monsoon and assists in assessing the ecological stress on riparian habitats during the peak summer contraction.

5.3 Modified Normalized Difference Water Index (MNDWI)

5.3.1 Definition and Scientific Significance

The Modified Normalized Difference Water Index (MNDWI) is an advanced remote sensing index developed to improve the extraction of open water features. While the standard NDWI is effective, it often struggles to separate water bodies from built-up areas and bare soil, which can reflect NIR light similarly to turbid water. By substituting the Near-Infrared (NIR) band with the Shortwave-Infrared (SWIR) band, MNDWI significantly enhances the spectral contrast between water and terrestrial noise. In the context of the Narmada Basin, MNDWI provides a highly accurate representation of the river's spatial extent, especially around the rocky and urbanized ghats of Omkareshwar and Maheshwar.

5.3.2 Mathematical Formulation

Water absorbs SWIR light even more strongly than NIR light, whereas built-up land and bare soil reflect SWIR light heavily. This fundamental spectral difference is captured using the Green and SWIR bands of the Sentinel-2 Level-2A imagery:

$$MNDWI = \frac{Band\ 3\ (Green) - Band\ 11\ (SWIR)}{Band\ 3\ (Green) + Band\ 11\ (SWIR)} \tag{5.3}$$

The resulting MNDWI values range from -1 to +1. Values greater than zero typically indicate pure water pixels, while negative values represent non-water surfaces.

5.3.3 Methodology and Application

The MNDWI computation was executed on the Google Earth Engine (GEE) platform. The methodology involved generating temporal composites to calculate the index for the defined stretches. By leveraging the SWIR band, this index successfully masked out the reflective sandbars, concrete ghat structures, and urban footprint surrounding the river, allowing for a highly precise extraction of the aquatic boundary for volume and flow analysis.

5.3.4 Temporal Analysis and Observations (January – May 2025)

The temporal variation in MNDWI provides a clear picture of the hydrological changes in the Narmada River leading up to the monsoon season.

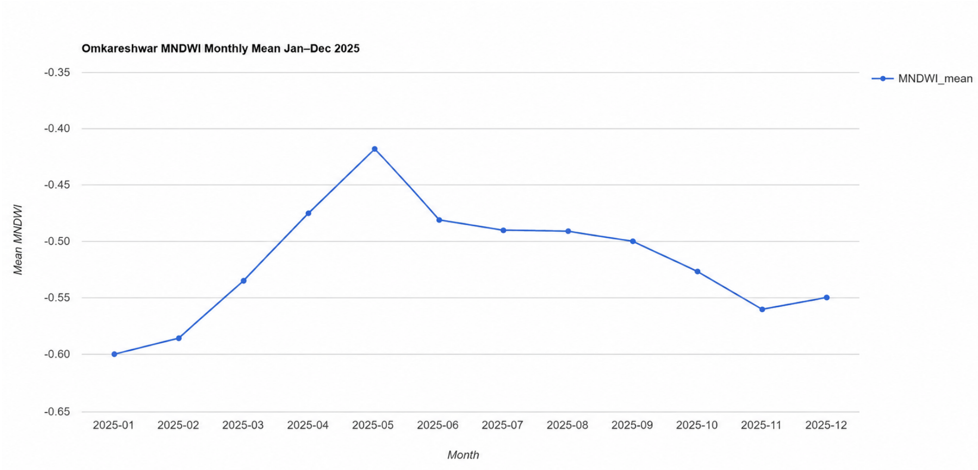


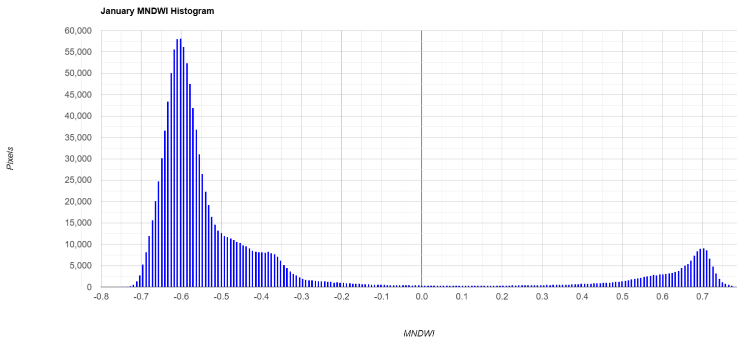
Figure 5.21: Monthly Mean MNDWI Trend for the Narmada River (Omkareshwar Stretch) for the year 2025.

As shown in Figure 5.21, the mean MNDWI value starts at its lowest point in January (approx. -0.60) and exhibits a steady, sharp incline through the pre-monsoon months, peaking in May (approx. -0.42). This upward trajectory suggests a relative increase in the detectable water signature or a reduction in exposed dry riverbed features due to scheduled dam releases upstream. The SWIR band's sensitivity highlights the subtle shifts in shallow water dynamics during the peak summer phase.

5.3.5 Monthly Histogram and Statistical Analysis

To evaluate the precise distribution of water versus non-water pixels, monthly histograms mapping the pixel frequency against MNDWI values were analyzed. The following subsections present the monthly visual distributions accompanied by their respective statistical data exported from GEE.

January 2025

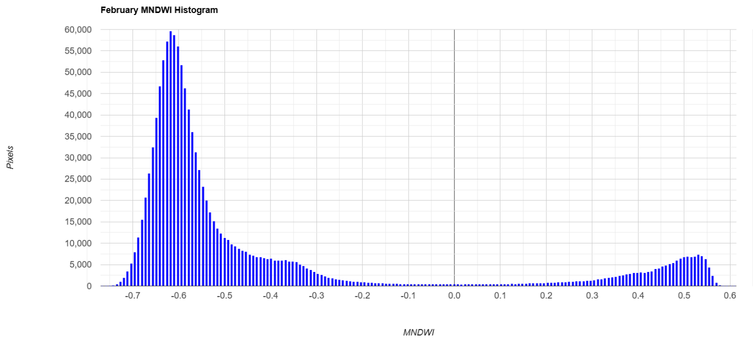


	A	B
1	Band Valu	MNDWI Count
2	-0.797	1
3	-0.789	
4	-0.781	
5	-0.773	
6	-0.766	
7	-0.758	5
8	-0.75	11
9	-0.742	28
10	-0.734	89
11	-0.727	228
12	-0.719	614
13	-0.711	1,419.81
14	-0.703	2,745.44
15	-0.695	5,297.34
16	-0.688	8,149.98
17	-0.68	11,994.82
18	-0.672	15,661.42

Figure 5.22: MNDWI pixel distribution histogram and statistical summary for January 2025.

The January histogram reveals a dominant peak in the negative spectrum (around -0.6), representing the extensive dry banks, exposed rocks, and dormant riparian zones. A distinct secondary peak is visible in the extreme positive range (near +0.7), which represents the pure, deep water of the central river channel during stable winter flows.

February 2025

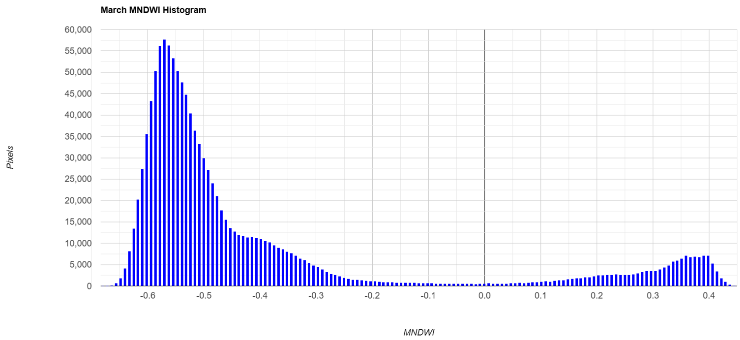


	A	B
1	Band Valu	MNDWI Count
2	-0.766	2
3	-0.758	12
4	-0.75	47
5	-0.742	141
6	-0.734	413
7	-0.727	1,026
8	-0.719	1,993.82
9	-0.711	3,420.98
10	-0.703	5,278.97
11	-0.695	7,913.37
12	-0.688	11,387.24
13	-0.68	15,533.69
14	-0.672	20,765.47
15	-0.664	26,356.74
16	-0.656	32,501.46
17	-0.648	39,367.79
18	-0.641	46,709.60

Figure 5.23: MNDWI pixel distribution histogram and statistical summary for February 2025.

In February, the distribution remains similar to January, but the positive peak indicating pure water broadens slightly. The statistical data confirms a low, stable mean MNDWI, reflecting undisturbed, low-velocity river conditions typical of late winter.

March 2025

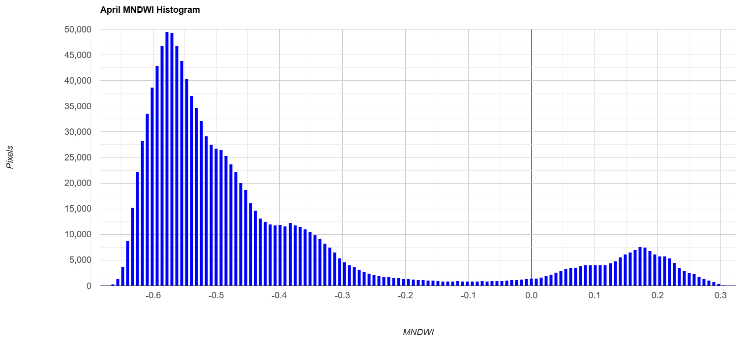


	A	B
1	Band Valu	MNDWI Count
2	-0.68	4
3	-0.672	32
4	-0.664	140
5	-0.656	640
6	-0.648	1,825
7	-0.641	4,154.74
8	-0.633	8,206.04
9	-0.625	13,520.70
10	-0.617	20,255.68
11	-0.609	27,429.93
12	-0.602	35,561.83
13	-0.594	43,301.95
14	-0.586	50,324.54
15	-0.578	56,152.41
16	-0.57	57,709.06
17	-0.563	56,283.17
18	-0.555	53,280.76

Figure 5.24: MNDWI pixel distribution histogram and statistical summary for March 2025.

The March data shows a progressive rightward shift of the negative peak, suggesting increased moisture in the riparian zones or a slight expansion of the water boundary. The positive pure-water peak (around +0.4) is highly distinct, marking the distinct separation of the river from its banks.

April 2025

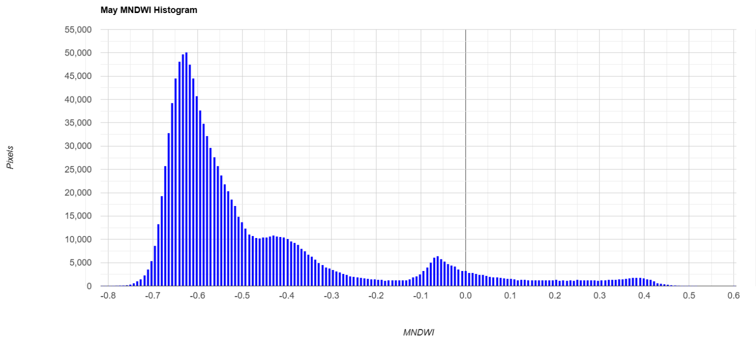


	A	B
1	Band Valu	MNDWI Count
2	-0.68	1
3	-0.672	40
4	-0.664	331
5	-0.656	1,343
6	-0.648	3,767.52
7	-0.641	8,762.82
8	-0.633	15,259.92
9	-0.625	22,127.95
10	-0.617	28,200.68
11	-0.609	33,551.02
12	-0.602	38,636.03
13	-0.594	42,927.48
14	-0.586	46,709.46
15	-0.578	49,489.46
16	-0.57	49,326.35
17	-0.563	46,807.91
18	-0.555	43,880.11

Figure 5.25: MNDWI pixel distribution histogram and statistical summary for April 2025.

During April, the histogram variance increases. While the dry land pixels dominate, the water pixels form a wider, lower-frequency hump in the positive domain. This indicates varying water depths and increased turbidity, which begins to affect the SWIR reflectance slightly as temperatures rise.

May 2025



	A	B
1	Band Valu	MNDWI Count
2	-0.813	1
3	-0.805	10
4	-0.797	17
5	-0.789	36
6	-0.781	73
7	-0.773	99
8	-0.766	118
9	-0.758	178
10	-0.75	359
11	-0.742	602
12	-0.734	1,052.97
13	-0.727	1,471.32
14	-0.719	2,290.65
15	-0.711	3,538.02
16	-0.703	5,352.90
17	-0.695	8,668.51
18	-0.688	13,314.52

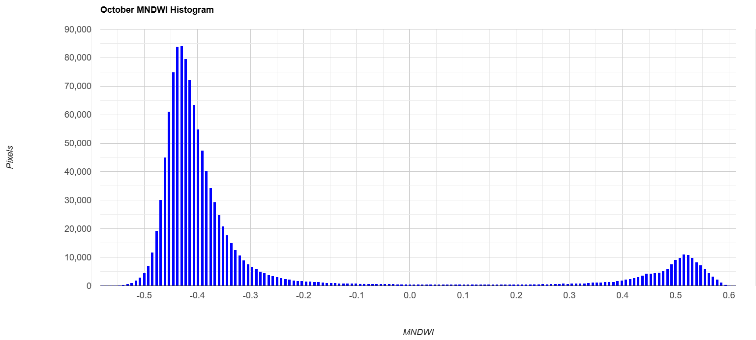
Figure 5.26: MNDWI pixel distribution histogram and statistical summary for May 2025.

May records the highest pre-monsoon mean MNDWI. The histogram exhibits a dense concentration of pixels shifting towards zero, with a small but concentrated peak in the positive range. This signifies maximum pre-monsoon water spread, largely sustained by controlled hydrological management in the basin despite peak summer evaporation.

5.3.6 Monsoon Data Exclusions (June – September)

The temporal dataset purposefully excludes the months of June, July, August, and September. During the peak Indian southwest monsoon, the Narmada Basin experiences dense, continuous cloud cover. Because MNDWI relies on the optical Shortwave-Infrared (SWIR) band, which cannot penetrate clouds, the resulting data for these months was heavily corrupted by atmospheric noise. Consequently, extracting reliable pixel histograms and CSV statistics during this four-month window was not viable, and the data was omitted to maintain the analytical integrity of the study.

October 2025

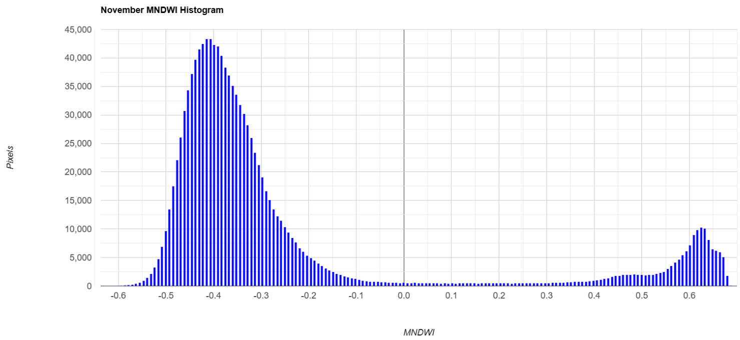


	A	B
1	Band Valu	MNDWI Count
2	-0.578	1
3	-0.57	2
4	-0.563	11
5	-0.555	35
6	-0.547	135
7	-0.539	333
8	-0.531	675.341
9	-0.523	1,108.69
10	-0.516	1,921.92
11	-0.508	2,940.05
12	-0.5	4,427.44
13	-0.492	7,167.34
14	-0.484	11,749.16
15	-0.477	19,364.33
16	-0.469	30,173.56
17	-0.461	45,065.02
18	-0.453	61,096.02

Figure 5.27: MNDWI pixel distribution histogram and statistical summary for October 2025.

Following the monsoon, the October histogram shows a massive spike in negative values combined with a defined positive peak near +0.5. The river volume is high, but the surrounding landscape is heavily saturated, causing complex spectral mixing. The CSV data reflects a post-monsoon normalization phase.

November 2025

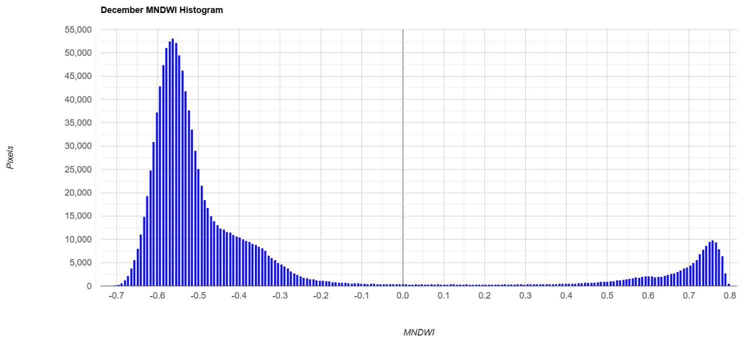


	A	B
1	Band Valu	MNDWI Count
2	-0.633	1
3	-0.625	
4	-0.617	7
5	-0.609	4
6	-0.602	27
7	-0.594	59
8	-0.586	118
9	-0.578	188
10	-0.57	269
11	-0.563	421
12	-0.555	594.761
13	-0.547	937.09
14	-0.539	1,435.34
15	-0.531	2,132.43
16	-0.523	3,256.98
17	-0.516	4,769.03
18	-0.508	6,876.30

Figure 5.28: MNDWI pixel distribution histogram and statistical summary for November 2025.

In November, the distribution begins to stabilize. The primary peak associated with terrestrial features narrows, and the positive water peak becomes exceptionally distinct (around +0.6). This indicates that floodwaters have receded, leaving a well-defined, clear river channel.

December 2025



	A	B
1	Band Valu	MNDWI Count
2	-0.734	5
3	-0.727	11
4	-0.719	23
5	-0.711	50
6	-0.703	153
7	-0.695	306.341
8	-0.688	656
9	-0.68	1,277.93
10	-0.672	2,262.87
11	-0.664	3,774.04
12	-0.656	5,592.98
13	-0.648	8,038.95
14	-0.641	11,084.46
15	-0.633	14,922.22
16	-0.625	19,282.05
17	-0.617	24,760.53
18	-0.609	30,935.68

Figure 5.29: MNDWI pixel distribution histogram and statistical summary for December 2025.

The December analysis marks a complete return to stable winter hydrology. The histogram displays a stark bimodal distribution with a very sharp peak in the high positive range (+0.7 to +0.8), representing exceptionally clear, deep water, clearly distinguished from the dry, exposed riverbanks.

5.3.7 Discussion and Ecological Implications

The application of the Modified Normalized Difference Water Index (MNDWI) provides a highly refined assessment of the Narmada River's spatial extent, successfully isolating the water body from the complex built-up and rocky environments of Omkareshwar and Maheshwar.

Ecologically, the distinct bimodal histograms observed in the winter months (January, February, and December) highlight stable, undisturbed aquatic habitats with clear boundaries. In contrast, the shifting distributions in the pre-monsoon (April-May) and post-monsoon (October) phases indicate significant morphological dynamism—where shallow banks, sandbars, and riparian zones alternate between exposure and inundation. Accurate mapping of these inundated zones using MNDWI is crucial. It allows environmental engineers to track the loss of critical riparian habitats during dry phases and map the true extent of floodplains following the monsoon, thereby supporting data-driven river governance and ecological preservation.

5.4 Automated Water Extraction Index (AWEI)

5.4.1 Definition and Scientific Significance

The Automated Water Extraction Index (AWEI) is a highly specialized remote sensing index formulated to improve the classification accuracy of water bodies in environments with complex topographies, severe shadowing, and dark urban surfaces. Unlike NDWI or MNDWI, which utilize normalized ratios, AWEI employs a multi-band coefficient approach to force non-water pixels into the negative numerical range while keeping water pixels positive or closer to zero. In the Narmada Basin, particularly around the structurally varied ghats of Omkareshwar and Maheshwar, AWEI is crucial for distinguishing genuine water surfaces from the dark topographical shadows cast by the steep riverbanks and architectural features.

5.4.2 Mathematical Formulation

AWEI utilizes a combination of the Green, Near-Infrared (NIR), and two Shortwave-Infrared (SWIR) bands. The non-shadow version ($AWEI_{nsh}$) is optimal for areas where urban shadows are not the primary noise factor, providing a stable extraction threshold:

$$AWEI_{nsh} = 4 \times (Band\ 3 - Band\ 11) - (0.25 \times Band\ 8 + 2.75 \times Band\ 12) \quad (5.4)$$

By aggressively penalizing the NIR and SWIR2 bands, the formula suppresses the spectral signatures of built-up areas and soil, leaving a clean map of the water body.

5.4.3 Methodology and Application

Processed within the Google Earth Engine (GEE) environment, the AWEI was applied to the Sentinel-2 imagery across the defined stretches. Because AWEI outputs are highly sensitive to sensor calibration, temporal median composites were utilized for each month. The index effectively isolated the Narmada River's main channel, suppressing false positives along the rocky, shadowy edges of the ghats and providing an accurate representation of the effective water spread.

5.4.4 Temporal Analysis and Observations (January – June 2025)

The temporal trend of the mean AWEI provides an excellent proxy for observing the river's spatial contraction during the critical dry season phase leading up to the monsoon.

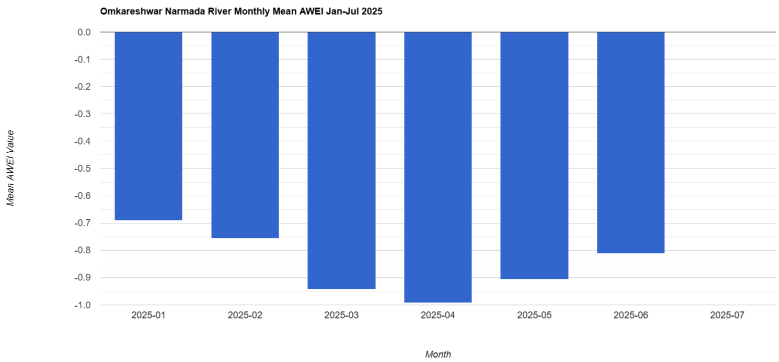


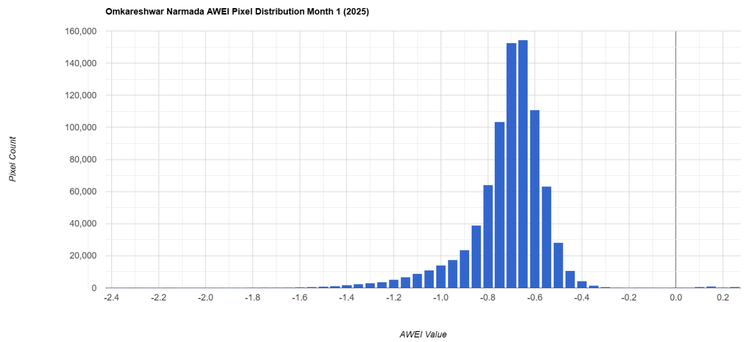
Figure 5.30: Monthly Mean AWEI Trend for the Narmada River (Omkareshwar Stretch) from January to June 2025.

Figure 5.30 illustrates a profound downward trajectory in the mean AWEI values from January through April. The mean starts at approximately -0.69 in January and drops severely to roughly -1.0 by April. Because AWEI pushes non-water pixels into deeper negative values, this sharp decline vividly captures the exposure of the riverbed and the expansion of dry, non-water surfaces as the river volume shrinks. A modest recovery is observed in May and June, where the values begin to climb back towards -0.8, reflecting early pre-monsoon flow alterations.

5.4.5 Monthly Histogram and Statistical Analysis

To map the precise distribution of these spectral signatures, monthly pixel frequency histograms were generated. The following sections detail the monthly visual distributions and their corresponding statistical summaries exported from GEE.

January 2025

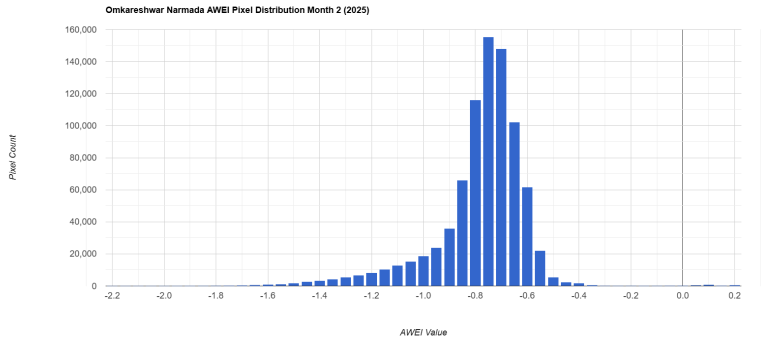


	A	B
1	Band Valu	AWEI Count
2	-2.4	1
3	-2.35	2
4	-2.3	
5	-2.25	1
6	-2.2	4
7	-2.15	
8	-2.1	7
9	-2.05	24
10	-2	8.733
11	-1.95	66.188
12	-1.9	78.463
13	-1.85	94.506
14	-1.8	193.725
15	-1.75	276.361
16	-1.7	268.11
17	-1.65	306.592
18	-1.6	452.796

Figure 5.31: AWEI pixel distribution histogram and statistical summary for January 2025.

The January histogram shows a distinct, relatively narrow bell-shaped distribution centered around -0.7. The statistical summary reflects stable winter flow conditions. The dominance of negative values indicates that the bounding box includes significant portions of the dry riparian zone alongside the river's main channel.

February 2025

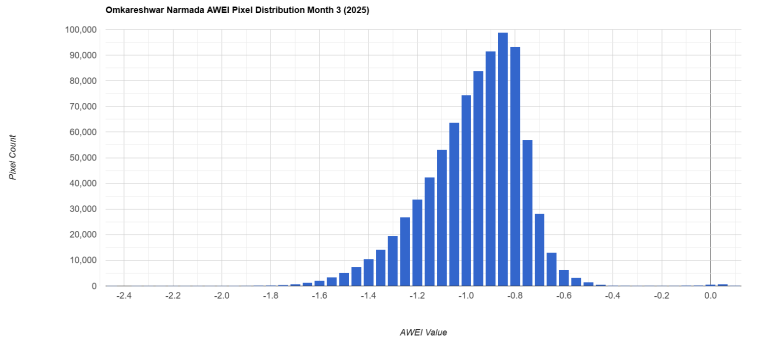


	A	B
1	Band Valu	AWEI Count
2	-2.2	13
3	-2.15	17
4	-2.1	19
5	-2.05	14
6	-2	26
7	-1.95	65
8	-1.9	112.6
9	-1.85	146.29
10	-1.8	207.282
11	-1.75	259.173
12	-1.7	413.969
13	-1.65	717.694
14	-1.6	1,072.70
15	-1.55	1,316.86
16	-1.5	1,849.45
17	-1.45	2,629.60
18	-1.4	3,257.48

Figure 5.32: AWEI pixel distribution histogram and statistical summary for February 2025.

In February, the histogram peak shifts slightly to the left, and the distribution remains sharp. The CSV statistics confirm a decrease in the mean AWEI, directly correlating with the natural post-winter recession of water levels and the subsequent initial exposure of riverbank sediments.

March 2025

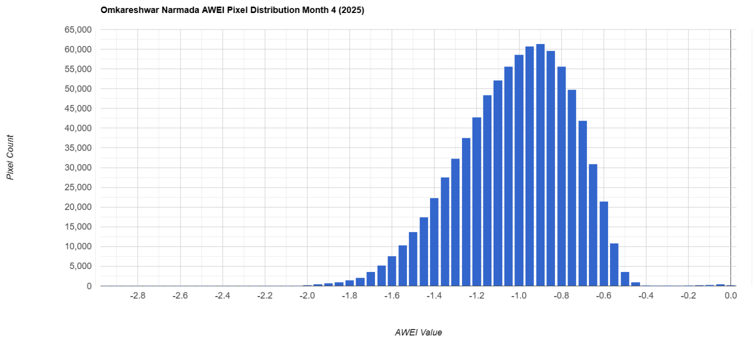


	A	B
1	Band Valu	AWEI Count
2	-2.45	1
3	-2.4	
4	-2.35	2
5	-2.3	6
6	-2.25	5
7	-2.2	3
8	-2.15	17.282
9	-2.1	13.318
10	-2.05	13
11	-2	19
12	-1.95	49
13	-1.9	102
14	-1.85	196.122
15	-1.8	245.714
16	-1.75	431.075
17	-1.7	684.11
18	-1.65	1,343.33

Figure 5.33: AWEI pixel distribution histogram and statistical summary for March 2025.

The March data illustrates a more aggressive leftward shift, with the peak moving past -0.8. This statistical drop represents the onset of the summer dry phase. The increased frequency of highly negative pixels confirms that shallow water zones are transitioning into exposed dry land.

April 2025

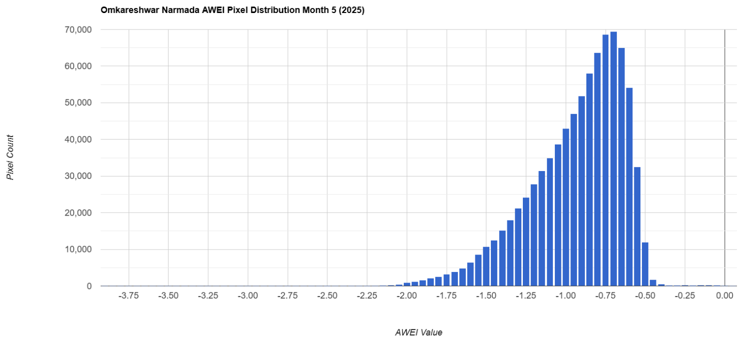


	A	B
1	Band Valu	AWEI Count
2	-2.95	1
3	-2.9	2
4	-2.85	7
5	-2.8	3
6	-2.75	4
7	-2.7	3
8	-2.65	1.282
9	-2.6	2
10	-2.55	4
11	-2.5	7
12	-2.45	10.318
13	-2.4	11
14	-2.35	12
15	-2.3	11
16	-2.25	12.8
17	-2.2	13
18	-2.15	15

Figure 5.34: AWEI pixel distribution histogram and statistical summary for April 2025.

April records the lowest mean AWEI of the observation period, with the histogram broadening and centering near the -1.0 mark. The statistical export reveals maximum variance, indicating peak summer contraction of the Narmada River, where topographical shadows and exposed rocky beds dominate the spectral response of the study area.

May 2025

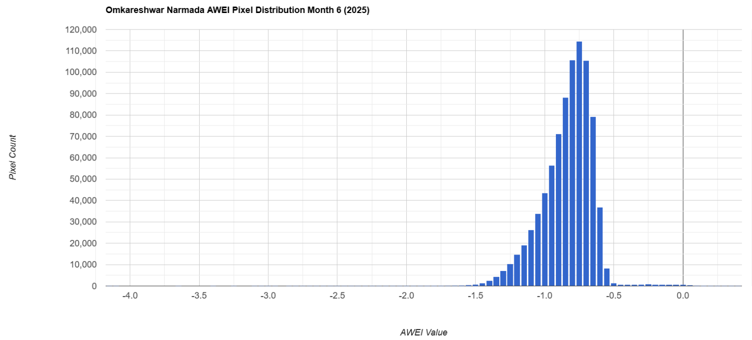


	A	B
1	Band Valu	AWEI Count
2	-3.9	1
3	-3.85	1.153
4	-3.8	1
5	-3.75	1
6	-3.7	1.561
7	-3.65	0.082
8	-3.6	0.286
9	-3.55	1
10	-3.5	3.314
11	-3.45	3
12	-3.4	8
13	-3.35	5
14	-3.3	2
15	-3.25	2
16	-3.2	2
17	-3.15	5
18	-3.1	4

Figure 5.35: AWEI pixel distribution histogram and statistical summary for May 2025.

In May, the trend reverses. The histogram exhibits a strong rightward shift, peaking sharply around -0.6 to -0.7 with a left-skewed tail. The CSV data confirms a recovery in the mean AWEI value. This suggests an influx of water—likely due to controlled dam releases regulating the basin's flow during peak summer demand.

June 2025



	A	B
1	Band Valu	AWEI Count
2	-4.15	0.004
3	-4.1	0.314
4	-4.05	
5	-4	
6	-3.95	
7	-3.9	
8	-3.85	
9	-3.8	
10	-3.75	
11	-3.7	
12	-3.65	0.282
13	-3.6	
14	-3.55	
15	-3.5	
16	-3.45	
17	-3.4	1
18	-3.35	

Figure 5.36: AWEI pixel distribution histogram and statistical summary for June 2025.

The June distribution remains heavily concentrated on the right side of the graph, similar to May, though slightly more dispersed. The statistical summary indicates increased instability in the index, acting as a precursor to the monsoon season as localized precipitation begins to alter the dry terrain's spectral signature.

5.4.6 Data Exclusions (July – December 2025)

For the AWEI analysis, the dataset is strictly constrained to the six-month window spanning January to June 2025. The months of July through December were excluded due to two primary factors. First, the Indian southwest monsoon (July–September) introduces persistent, optically impenetrable cloud cover over the Narmada Basin, rendering multi-band indices like AWEI computationally invalid during this period. Second, the fundamental objective of applying AWEI in this study is to map the river's spatial contraction and shallow-water dynamics during the dry season. Once the post-monsoon floodplains are established, indices like MNDWI are sufficient, making the continued computation of AWEI redundant for the latter half of the year.

5.4.7 Discussion and Ecological Implications

The temporal application of the Automated Water Extraction Index (AWEI) provides a highly resilient map of the Narmada River's surface area, exceptionally well-filtered against the topographical noise of the Omkareshwar and Maheshwar ghats.

From an ecological standpoint, the severe drop in AWEI values from January to April starkly quantifies the loss of aquatic habitat area. As the river recedes and the index becomes highly negative, it signals the extensive exposure of the riverbed, which can disrupt the breeding grounds of local aquatic species and lead to the desiccation of shallow-water flora. The subsequent recovery in May highlights the artificial hydrodynamic regulation of the basin. Monitoring these exact AWEI thresholds allows for a highly accurate, noise-free assessment of ecological stress limits during the peak of the dry season, providing vital data for sustainable river basin governance.

5.5 Floating Algae Index (FAI)

5.5.1 Definition and Scientific Significance

The Floating Algae Index (FAI) is a specialized remote sensing metric designed to detect and quantify floating algae, phytoplankton, and surface aquatic vegetation. Unlike traditional vegetation indices such as NDVI, FAI is highly resistant to atmospheric interference, sunglint, and variations in water background reflectance. In the context of the Narmada Basin, particularly around the Omkareshwar and Maheshwar stretches, FAI is critical for monitoring eutrophication, identifying harmful algal blooms (HABs), and assessing the overall ecological balance and oxygen availability within the river ecosystem.

5.5.2 Mathematical Formulation

FAI is calculated using a baseline subtraction method across the visible, Near-Infrared (NIR), and Shortwave-Infrared (SWIR) spectrums. By linearly interpolating the baseline reflectance between the Red and SWIR bands, FAI measures the difference between the actual NIR reflectance and the expected baseline. Using Sentinel-2 Level-2A imagery, the formula is defined as:

$$FAI = \rho_{NIR} - \left[\rho_{Red} + (\rho_{SWIR1} - \rho_{Red}) \times \frac{\lambda_{NIR} - \lambda_{Red}}{\lambda_{SWIR1} - \lambda_{Red}}\right] \tag{5.5}$$

Where ρ represents the surface reflectance of the respective bands (Band 8 for NIR, Band 4 for Red, and Band 11 for SWIR1), and λ represents their central wavelengths. Positive FAI values indicate the presence of floating algae, with higher values corresponding to denser algal blooms.

5.5.3 Methodology and Application

The FAI was computed within the Google Earth Engine (GEE) cloud platform. To prevent false positives from terrestrial riparian vegetation, a rigorous water mask (derived from MNDWI) was applied first, restricting the FAI calculation strictly to the river's surface. Temporal median composites were generated to minimize transient noise, allowing for the precise spatial mapping of algal concentrations resulting from agricultural runoff, seasonal stagnation, and temperature fluctuations.

5.5.4 Temporal Analysis and Observations

The monthly progression of FAI reveals critical insights into the biological response of the Narmada River to seasonal climatic variations and nutrient loading.

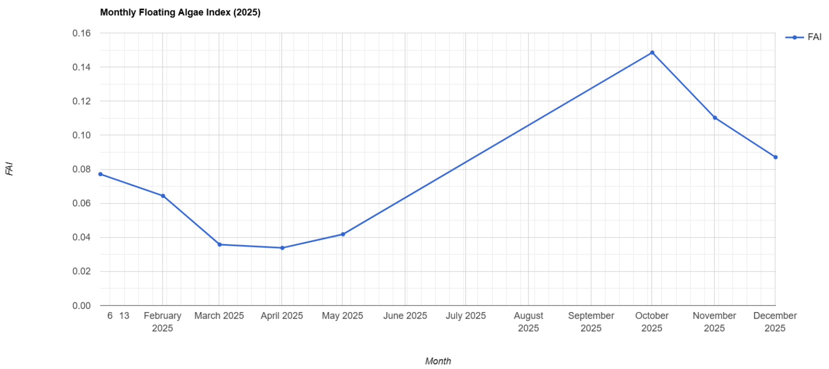


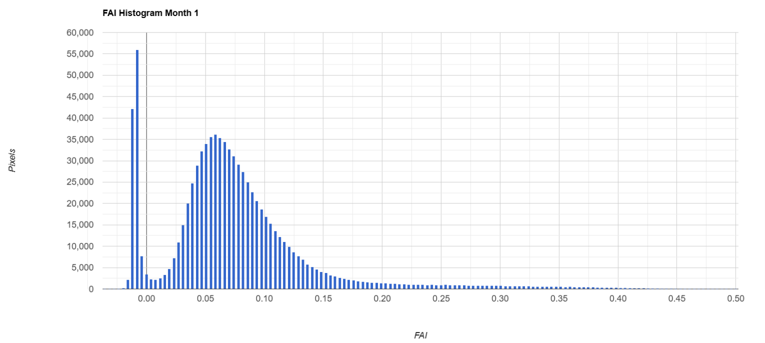
Figure 5.37: Monthly Mean FAI Trend for the Narmada River (Omkareshwar Stretch) for the year 2025.

As depicted in Figure 5.37, the mean FAI displays a highly dynamic seasonal curve. In January and February, the index maintains a moderate level (approx. 0.07 to 0.06), which subsequently drops to its lowest points in March and April (near 0.035). This pre-summer decline suggests minimal algal proliferation due to stable flow and lower nutrient concentrations. However, a slight rise begins in May. Following the monsoon season (which is excluded from this specific chart), October exhibits a massive spike in FAI (reaching nearly 0.15), indicating severe post-monsoon algal blooms triggered by nutrient-rich agricultural runoff. The trend then sharply declines through November and December as temperatures drop and nutrients deplete.

5.5.5 Monthly Histogram and Statistical Analysis

To quantify the pixel-level distribution of floating algae across the study area, month-by-month histograms were generated. These graphs map the frequency of pixels against their FAI values. The following subsections present the monthly visual distributions accompanied by their respective statistical data exported directly from GEE.

January 2025

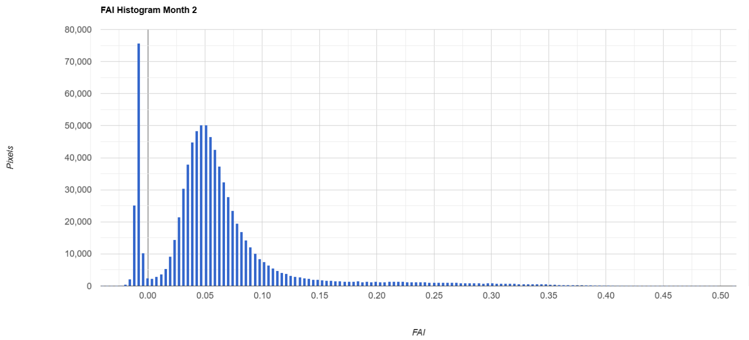


	A	B
1	Band Valu	FAI Count
2	-0.035	1
3	-0.031	1
4	-0.027	3
5	-0.023	29
6	-0.02	224
7	-0.016	2,151.38
8	-0.012	42,154.75
9	-0.008	55,990.04
10	-0.004	7,660.90
11	0	3,401.53
12	0.004	2,285.14
13	0.008	2,236.14
14	0.012	2,488.81
15	0.016	3,310.88
16	0.02	4,726.18
17	0.023	7,287.99
18	0.027	10,924.94

Figure 5.38: FAI pixel distribution histogram and statistical summary for January 2025.

The January histogram shows a wide distribution with a notable peak in the low positive range. This indicates a moderate baseline presence of floating aquatic vegetation and residual winter algae. The statistical summary confirms a moderate mean FAI, typical of stable winter hydrology where biological activity is sustained but not explosive.

February 2025

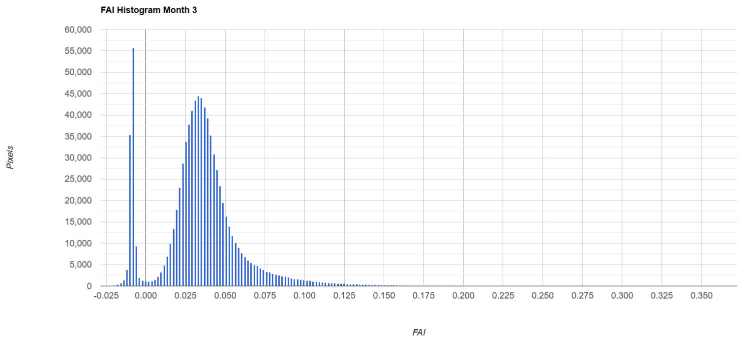


	A	B
1	Band Valu	FAI Count
2	-0.039	1
3	-0.035	4
4	-0.031	7
5	-0.027	30
6	-0.023	107
7	-0.02	503.2
8	-0.016	2,224.91
9	-0.012	25,171.35
10	-0.008	75,758.30
11	-0.004	10,276.59
12	0	2,422.47
13	0.004	2,254.25
14	0.008	2,856.53
15	0.012	3,676.90
16	0.016	5,423.85
17	0.02	9,158.72
18	0.023	14,374.70

Figure 5.39: FAI pixel distribution histogram and statistical summary for February 2025.

In February, the histogram shifts slightly to the left, indicating a reduction in overall algal density. The CSV statistics corroborate a decrease in the mean FAI, reflecting clearer surface water conditions as temperatures begin to slowly shift before the onset of the extreme summer heat.

March 2025

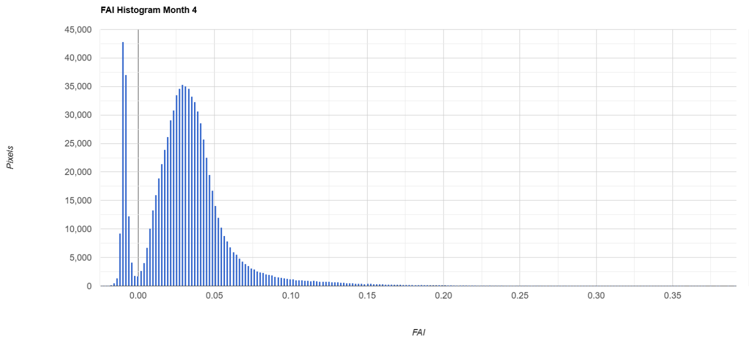


	A	B
1	Band Valu	FAI Count
2	-0.027	5
3	-0.025	3
4	-0.023	13
5	-0.021	30.106
6	-0.02	101
7	-0.018	326
8	-0.016	738.612
9	-0.014	1,373.68
10	-0.012	3,839.38
11	-0.01	35,315.12
12	-0.008	55,777.87
13	-0.006	9,272.18
14	-0.004	1,993.82
15	-0.002	1,256.24
16	0	1,124.05
17	0.002	1,065.78
18	0.004	1,173.18

Figure 5.40: FAI pixel distribution histogram and statistical summary for March 2025.

The March data illustrates a pronounced decline in FAI values, with the histogram concentrating heavily towards the zero mark. This statistical drop represents the lowest biological activity on the river surface, likely due to increased flow regulation or nutrient dilution during this specific seasonal window.

April 2025

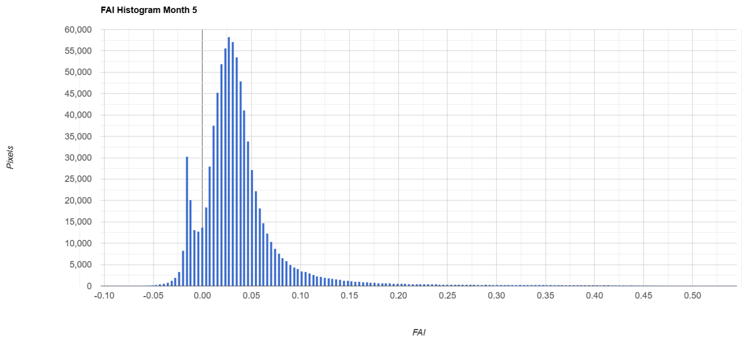


	A	B
1	Band Valu	FAI Count
2	-0.023	1
3	-0.021	14.906
4	-0.02	36
5	-0.018	151
6	-0.016	522.2
7	-0.014	1,379.77
8	-0.012	9,240.89
9	-0.01	42,802.02
10	-0.008	37,012.95
11	-0.006	12,236.54
12	-0.004	4,156.28
13	-0.002	1,794.50
14	0	1,722.60
15	0.002	2,634.59
16	0.004	4,101.68
17	0.006	6,694.73
18	0.008	10,148.70

Figure 5.41: FAI pixel distribution histogram and statistical summary for April 2025.

During April, the mean FAI hits its lowest point of the year. The histogram is tightly packed near the lower threshold, confirming minimal floating algae. This suggests that despite rising temperatures, the necessary nutrient loads required for algal blooms are absent or diluted in the main channel.

May 2025



	A	B
1	Band Valu	FAI Count
2	-0.102	2
3	-0.098	
4	-0.094	1
5	-0.09	1
6	-0.086	2
7	-0.082	7
8	-0.078	7
9	-0.074	7
10	-0.07	15
11	-0.066	32
12	-0.063	41
13	-0.059	72
14	-0.055	113.969
15	-0.051	169.925
16	-0.047	272.125
17	-0.043	416.694
18	-0.039	550.047

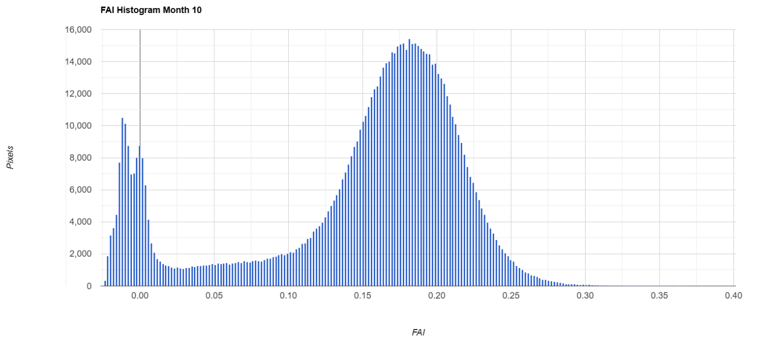
Figure 5.42: FAI pixel distribution histogram and statistical summary for May 2025.

In May, the histogram begins to widen and shift to the right, indicating a resurgence in algal growth. The statistical export reveals an increasing mean FAI. High summer temperatures combined with localized water stagnation along the ghat edges create optimal breeding conditions for early phytoplankton proliferation.

5.5.6 Monsoon Data Exclusions (June – September)

The data spanning June, July, August, and September has been explicitly excluded from this analysis. The FAI relies heavily on the optical Near-Infrared (NIR) and Shortwave-Infrared (SWIR) bands. During the Indian southwest monsoon, the Narmada Basin is blanketed by heavy, continuous cloud cover, rendering optical sensor data statistically unreliable and prone to extreme atmospheric noise. Consequently, extracting valid FAI histograms during these four months is scientifically unviable.

October 2025

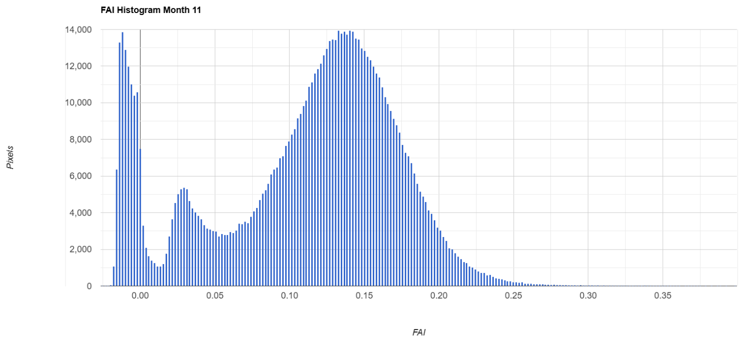


	A	B
1	Band Valu	FAI Count
2	-0.025	4
3	-0.023	330.945
4	-0.021	1,860.33
5	-0.02	3,174.42
6	-0.018	3,610.64
7	-0.016	4,456.37
8	-0.014	7,698.21
9	-0.012	10,500.59
10	-0.01	10,145.67
11	-0.008	8,754.14
12	-0.006	6,973.02
13	-0.004	7,018.42
14	-0.002	8,028.20
15	0	8,756.95
16	0.002	7,983.52
17	0.004	6,295.68
18	0.006	4,151.78

Figure 5.43: FAI pixel distribution histogram and statistical summary for October 2025.

Following the monsoon, the October histogram reveals a dramatic rightward shift, showcasing the highest FAI values of the year. The river experiences massive post-monsoon eutrophication driven by the influx of nutrient-dense agricultural runoff. The CSV data confirms a severe peak in mean FAI, reflecting widespread algal blooms across the stretches.

November 2025

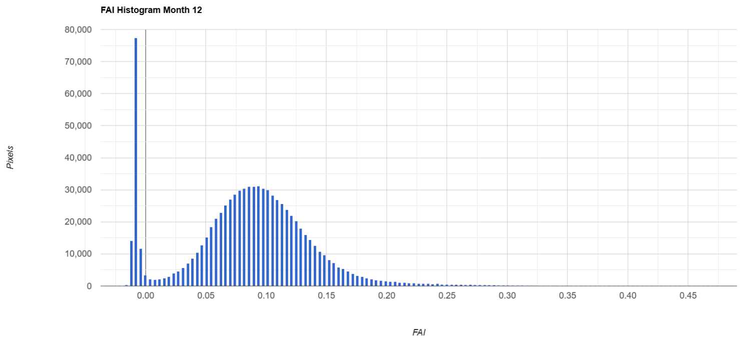


	A	B
1	Band Valu	FAI Count
2	-0.025	1
3	-0.023	1
4	-0.021	5
5	-0.02	51
6	-0.018	1,063.17
7	-0.016	6,360.69
8	-0.014	13,302.08
9	-0.012	13,862.14
10	-0.01	12,885.22
11	-0.008	11,992.72
12	-0.006	11,027.27
13	-0.004	10,404.43
14	-0.002	10,584.48
15	0	7,510.33
16	0.002	3,317.59
17	0.004	2,089.90
18	0.006	1,628.37

Figure 5.44: FAI pixel distribution histogram and statistical summary for November 2025.

In November, the distribution begins to recede as post-monsoon nutrient levels are gradually exhausted. While the histogram still shows a significant presence of algae, the statistical data indicates a sharp decline from the October peak, marking the initial recovery phase of the river's surface clarity.

December 2025



	A	B
1	Band Valu	FAI Count
2	-0.035	1
3	-0.031	
4	-0.027	
5	-0.023	7
6	-0.02	50
7	-0.016	340.906
8	-0.012	14,134.37
9	-0.008	77,341.17
10	-0.004	11,593.15
11	0	3,317.78
12	0.004	2,074.42
13	0.008	2,040.71
14	0.012	2,192.98
15	0.016	2,513.25
16	0.02	2,949.22
17	0.023	3,933.02
18	0.027	4,655.69

Figure 5.45: FAI pixel distribution histogram and statistical summary for December 2025.

The December analysis marks a return toward winter baseline conditions. The histogram shifts back towards the lower positive range, and the CSV statistics demonstrate a stabilized, reduced mean FAI. Cooler temperatures and diminished surface runoff effectively halt further algal proliferation.

5.5.7 Discussion and Ecological Implications

The temporal analysis of the Floating Algae Index (FAI) provides a critical biological health indicator for the Narmada Basin. The drastic fluctuations observed—from the dormant phases in March and April to the extreme eutrophic spike in October—highlight the profound impact of seasonal agricultural runoff on the river's ecological balance.

Ecologically, the massive post-monsoon algal blooms (as captured in the October data) pose a significant threat to aquatic health. Densely packed floating algae block sunlight from reaching benthic vegetation and, upon decomposition, drastically deplete dissolved oxygen (DO) levels in the water column, potentially leading to localized hypoxic dead zones. Continuous monitoring of these FAI peaks is crucial. It empowers environmental agencies to pinpoint exact pollution entry points along the Omkareshwar and Maheshwar stretches and deploy preemptive bioremediation strategies before critical oxygen depletion threatens the local aquatic fauna.

5.6 Normalized Difference Vegetation Index (NDVI)

5.6.1 Definition and Scientific Significance

The Normalized Difference Vegetation Index (NDVI) is the most widely utilized remote sensing metric for assessing the presence, density, and health of live green vegetation. While primarily an agricultural and forestry metric, its application in river basin monitoring is highly significant for evaluating the ecological health of riparian zones. In the Narmada Basin, tracking NDVI along the Omkareshwar and Maheshwar stretches allows researchers to monitor the seasonal dynamics of riverbank vegetation, detect the emergence of densely packed surface flora, and assess the ecological degradation or vitality of the immediate watershed environment.

5.6.2 Mathematical Formulation

NDVI is predicated on the physiological characteristics of healthy vegetation, which strongly absorbs Red light for photosynthesis while highly reflecting Near-Infrared (NIR) light due to its internal cellular structure. Using Sentinel-2 Level-2A imagery, the index is computed as the normalized ratio between these two bands:

$$NDVI = \frac{Band\ 8\ (NIR) - Band\ 4\ (Red)}{Band\ 8\ (NIR) + Band\ 4\ (Red)} \tag{5.6}$$

The resulting NDVI values range from -1 to +1. Negative values generally correspond to water bodies or highly saturated soils. Values around zero indicate bare soil or rock, whereas positive values (typically 0.2 to 0.8) indicate varying densities of healthy, green vegetation.

5.6.3 Methodology and Application

The NDVI was computed utilizing the cloud-processing capabilities of Google Earth Engine (GEE). A spatial boundary encompassing the river channel and its immediate riparian buffer was defined. By analyzing this specific corridor, the index effectively captures the seasonal senescence (drying) and post-monsoon greening of the riverbanks. Temporal median composites were utilized to ensure that transient atmospheric noise did not artificially inflate or deflate the vegetation health metrics.

5.6.4 Temporal Analysis and Observations

The temporal trend of mean NDVI serves as a precise indicator of the seasonal biological cycle of the vegetation surrounding the river.

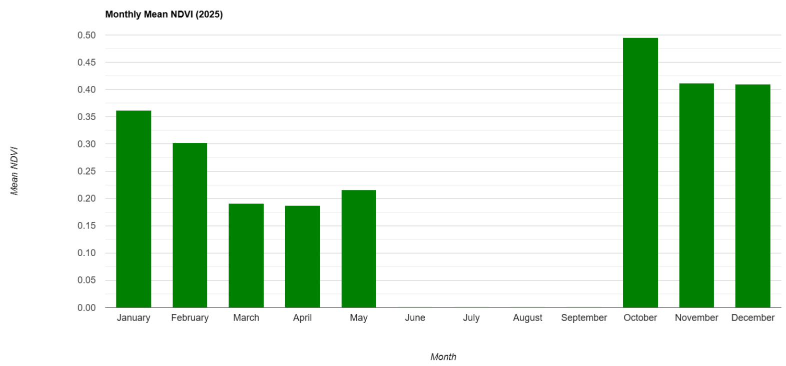


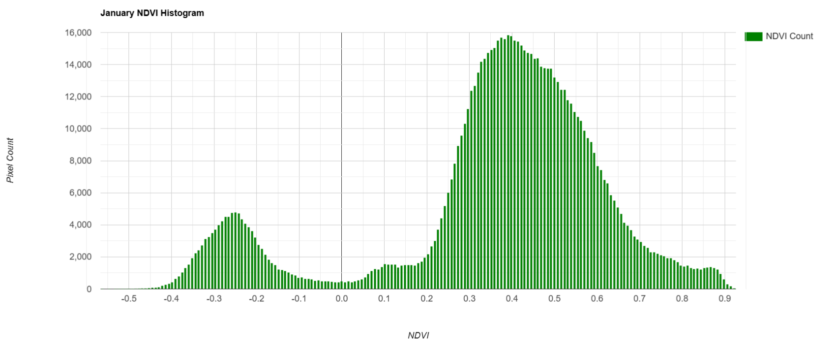
Figure 5.46: Monthly Mean NDVI Trend for the Narmada River (Omkareshwar Stretch) for the year 2025.

Figure 5.46 illustrates a pronounced seasonal vegetation cycle. At the beginning of the year (January and February), the mean NDVI remains moderately high (0.30 to 0.36), representing robust winter vegetation. As the region transitions into the dry summer phase, the index drops sharply, hitting its lowest points in March (≈ 0.19) and April (≈ 0.18). This decline marks the severe desiccation of riparian flora and the exposure of dry, non-vegetated riverbanks. Post-monsoon, the October data reveals an explosive peak in NDVI (approaching 0.50), showcasing a massive flush of new, dense green vegetation driven by monsoon saturation.

5.6.5 Monthly Histogram and Statistical Analysis

To comprehensively map the distribution of vegetated versus non-vegetated pixels within the study area, month-by-month histograms were generated. The following sections detail the pixel frequency distributions and their corresponding statistical summaries exported from GEE.

January 2025

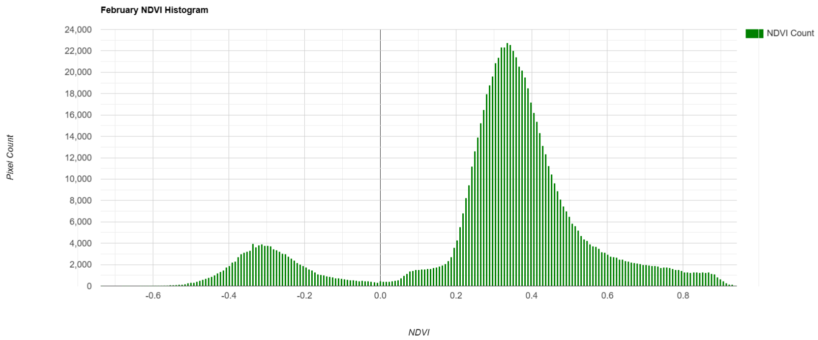


	A	B
1	Band Valu	NDVI Count
2	-0.563	1
3	-0.555	
4	-0.547	1
5	-0.539	
6	-0.531	2
7	-0.523	7
8	-0.516	6
9	-0.508	9
10	-0.5	12
11	-0.492	10
12	-0.484	18
13	-0.477	25
14	-0.469	35
15	-0.461	39
16	-0.453	63
17	-0.445	91.749
18	-0.438	102

Figure 5.47: NDVI pixel distribution histogram and statistical summary for January 2025.

The January histogram exhibits a bimodal distribution. The smaller peak in the negative range represents the water body, while the dominant, broad peak in the high positive range (0.3 to 0.6) represents healthy winter riparian vegetation. The statistical summary confirms a strong vegetative presence across the studied banks.

February 2025

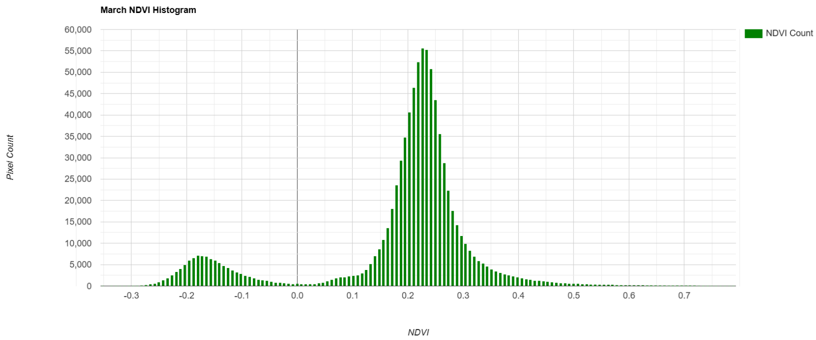


	A	B
1	Band Valu	NDVI Count
2	-0.734	1
3	-0.727	
4	-0.719	
5	-0.711	
6	-0.703	
7	-0.695	
8	-0.688	
9	-0.68	1
10	-0.672	0.922
11	-0.664	1
12	-0.656	
13	-0.648	
14	-0.641	
15	-0.633	2
16	-0.625	2
17	-0.617	7
18	-0.609	9

Figure 5.48: NDVI pixel distribution histogram and statistical summary for February 2025.

In February, the bimodal structure remains, but the positive vegetation peak begins to shift slightly to the left, indicating the early stages of declining chlorophyll activity as temperatures begin to rise and soil moisture slightly decreases.

March 2025

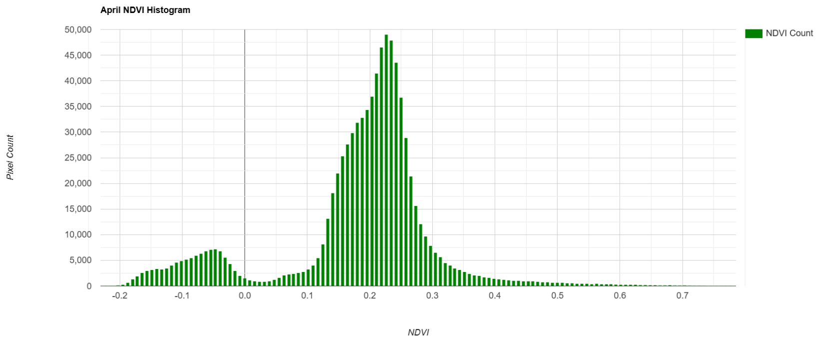


	A	B
1	Band Valu	NDVI Count
2	-0.352	1
3	-0.344	
4	-0.336	1
5	-0.328	2
6	-0.32	1
7	-0.313	1
8	-0.305	3
9	-0.297	22
10	-0.289	46
11	-0.281	130
12	-0.273	254
13	-0.266	420
14	-0.258	612.396
15	-0.25	922.114
16	-0.242	1,328.53
17	-0.234	1,858.99
18	-0.227	2,576.43

Figure 5.49: NDVI pixel distribution histogram and statistical summary for March 2025.

The March data illustrates a severe decline in vegetation health. The histogram's primary peak compresses heavily towards the lower positive range (0.2), confirming widespread senescence. The CSV export reflects a sharp drop in mean NDVI, aligning with the peak dry season where only resilient or heavily irrigated flora survives.

April 2025

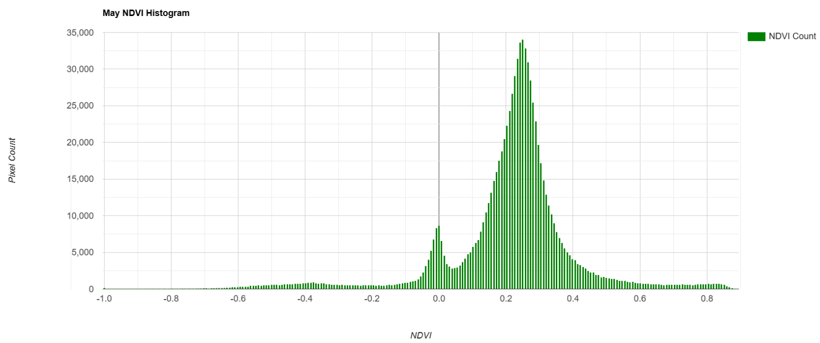


	A	B
1	Band Valu	NDVI Count
2	-0.227	4
3	-0.219	11
4	-0.211	39
5	-0.203	127.678
6	-0.195	298.682
7	-0.188	710.592
8	-0.18	1,326.23
9	-0.172	1,924.16
10	-0.164	2,635.69
11	-0.156	2,952.84
12	-0.148	3,195.66
13	-0.141	3,312.29
14	-0.133	3,223.14
15	-0.125	3,495.91
16	-0.117	4,064.46
17	-0.109	4,635.06
18	-0.102	4,925.28

Figure 5.50: NDVI pixel distribution histogram and statistical summary for April 2025.

April records the lowest vegetative vitality. The histogram's right-side distribution thins out significantly, concentrating heavily around 0.1 to 0.2. This denotes maximum exposure of bare soil and dry rocks along the ghats, alongside highly stressed riparian vegetation due to extreme summer heat.

May 2025



	A	B
1	Band Valu	NDVI Count
2	-1	155.235
3	-0.992	4.173
4	-0.984	30
5	-0.977	2.965
6	-0.969	13.204
7	-0.961	5
8	-0.953	7.051
9	-0.945	8.859
10	-0.938	12
11	-0.93	23
12	-0.922	23
13	-0.914	15
14	-0.906	11.463
15	-0.898	23.435
16	-0.891	20.933
17	-0.883	18.62
18	-0.875	25.533

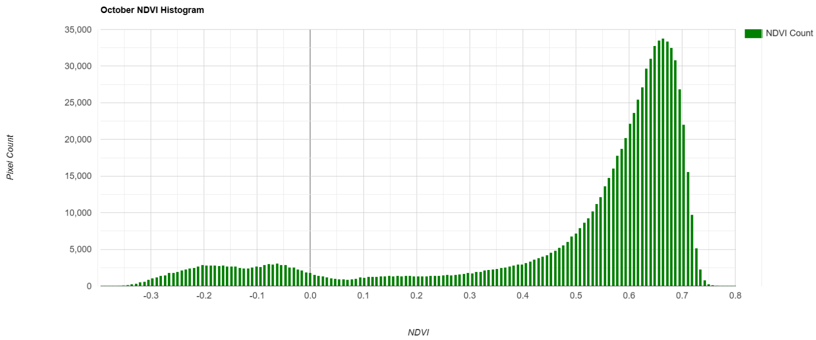
Figure 5.51: NDVI pixel distribution histogram and statistical summary for May 2025.

In May, the histogram shows a slight recovery, with the positive peak broadening marginally compared to April. This minor resurgence in the mean NDVI, as seen in the statistical summary, may be attributed to early localized pre-monsoon showers or specific resilient shrub species reacting to temperature shifts.

5.6.6 Monsoon Data Exclusions (June – September)

The dataset explicitly omits the months of June, July, August, and September. During the Indian southwest monsoon, persistent and dense cloud cover completely obscures optical satellite sensors. Since NDVI requires clear measurements of Red and Near-Infrared surface reflectance, computing the index during this four-month period would yield statistically corrupt and ecologically misleading data. Hence, these months are excluded to preserve analytical integrity.

October 2025

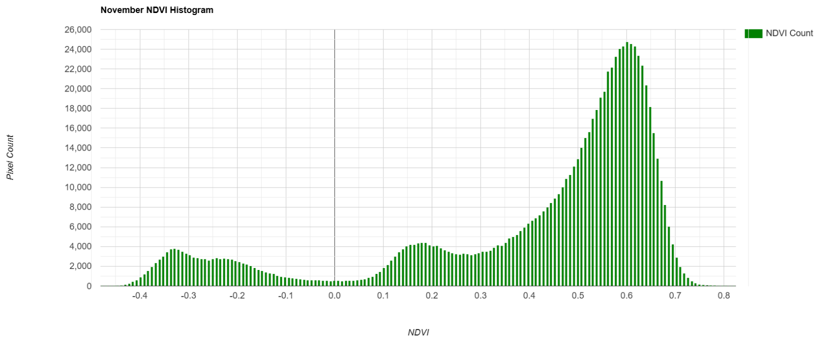


	A	B
1	Band Valu	NDVI Count
2	-0.391	5
3	-0.383	4
4	-0.375	10
5	-0.367	22
6	-0.359	33
7	-0.352	82
8	-0.344	151
9	-0.336	253
10	-0.328	357
11	-0.32	527
12	-0.313	613.945
13	-0.305	873
14	-0.297	1,060.02
15	-0.289	1,222.13
16	-0.281	1,410.54
17	-0.273	1,472.39
18	-0.266	1,806.62

Figure 5.52: NDVI pixel distribution histogram and statistical summary for October 2025.

Following the monsoon, the October histogram demonstrates an explosive shift to the right. A massive, dense peak forms between 0.6 and 0.7, representing the highest greenness of the year. The CSV data confirms a peak mean NDVI, reflecting a completely saturated riparian zone flush with dense, highly photosynthesizing post-monsoon vegetation.

November 2025

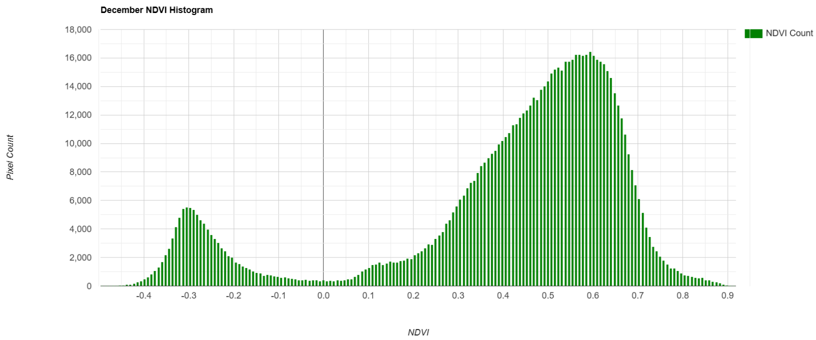


	A	B
1	Band Valu	NDVI Count
2	-0.477	1
3	-0.469	3
4	-0.461	7
5	-0.453	13
6	-0.445	28
7	-0.438	53
8	-0.43	132
9	-0.422	273
10	-0.414	428
11	-0.406	615
12	-0.398	912
13	-0.391	1,198
14	-0.383	1,550.32
15	-0.375	1,960.04
16	-0.367	2,323.91
17	-0.359	2,697.16
18	-0.352	3,013.91

Figure 5.53: NDVI pixel distribution histogram and statistical summary for November 2025.

In November, the distribution remains heavily weighted in the high positive domain but shows early signs of contraction compared to the October peak. As post-monsoon surface moisture begins to evaporate, the extremely high chlorophyll activity stabilizes, moving toward standard winter levels.

December 2025



	A	B
1	Band Valu	NDVI Count
2	-0.492	1
3	-0.484	4
4	-0.477	5
5	-0.469	11
6	-0.461	12
7	-0.453	32
8	-0.445	35
9	-0.438	94
10	-0.43	97
11	-0.422	159
12	-0.414	271
13	-0.406	339
14	-0.398	476
15	-0.391	621.847
16	-0.383	839.016
17	-0.375	1,072.71
18	-0.367	1,315.33

Figure 5.54: NDVI pixel distribution histogram and statistical summary for December 2025.

The December analysis marks the establishment of the winter baseline. The histogram retains a strong bimodal shape with a healthy vegetation peak around 0.5 to 0.6. The statistical data confirms a stable, highly vegetated riparian environment, closing the annual ecological cycle of the Narmada Basin stretches.

5.6.7 Discussion and Ecological Implications

The 12-month temporal monitoring of the Normalized Difference Vegetation Index (NDVI) clearly demarcates the severe seasonal extremes experienced by the Narmada Basin's riparian ecosystem.

Ecologically, the stark contrast between the April low and the October high highlights the ecosystem's heavy reliance on the monsoon cycle. During the pre-monsoon dry phase (March-April), the sharp decline in NDVI signifies severe moisture stress, which can compromise riverbank stability by weakening the root structures that prevent soil erosion. Conversely, the dense vegetative flush observed in October acts as a natural buffer, stabilizing the banks against post-monsoon flow velocities and filtering terrestrial runoff before it enters the river. Continuous tracking of these NDVI parameters provides an essential framework for identifying zones vulnerable to erosion during the dry season and ensuring proactive basin conservation.

5.7 Bare Soil Index (BSI)

5.7.1 Definition and Scientific Significance

The Bare Soil Index (BSI) is a numerical indicator used to reliably detect and quantify the presence of bare soil, exposed rocks, and non-vegetated surfaces. In river basin analysis, BSI is a highly significant metric for monitoring riverbank erosion, mapping sediment deposition, and tracking the seasonal exposure of sandbars within the river channel. By differentiating bare soil from water bodies and vegetative cover, BSI helps in evaluating the morphological dynamism of the Omkareshwar and Maheshwar stretches, particularly during the severe contraction of the river in the pre-monsoon summer months.

5.7.2 Mathematical Formulation

BSI leverages a combination of four distinct spectral bands: Shortwave-Infrared (SWIR), Red, Near-Infrared (NIR), and Blue. Bare soil exhibits high reflectance in the SWIR and Red spectrums, while vegetation and water strongly reflect in the NIR and Blue spectrums, respectively. Utilizing Sentinel-2 Level-2A surface reflectance data, the index is mathematically formulated as:

$$BSI = \frac{(Band\ 11\ (SWIR1) + Band\ 4\ (Red)) - (Band\ 8\ (NIR) + Band\ 2\ (Blue))}{(Band\ 11\ (SWIR1) + Band\ 4\ (Red)) + (Band\ 8\ (NIR) + Band\ 2\ (Blue))} \tag{5.7}$$

The resulting BSI values span from -1 to +1. High positive values denote bare soil exposure and sedimentation, whereas zero or negative values indicate the presence of vegetation, water, or high moisture content.

5.7.3 Methodology and Application

The calculation of BSI was executed within Google Earth Engine (GEE). A spatial filter was applied to restrict the analysis to the immediate river channel and its adjacent riparian boundaries. Temporal median composites were generated to eliminate cloud shadows and short-term anomalies. The primary application of BSI in this study is to visually and statistically track the desiccation of the riverbanks and the emergence of dry, exposed riverbed features as the water levels recede throughout the dry season.

5.7.4 Temporal Analysis and Observations

The month-to-month variation in the mean Bare Soil Index provides a direct geospatial measurement of seasonal land exposure within the Narmada Basin.

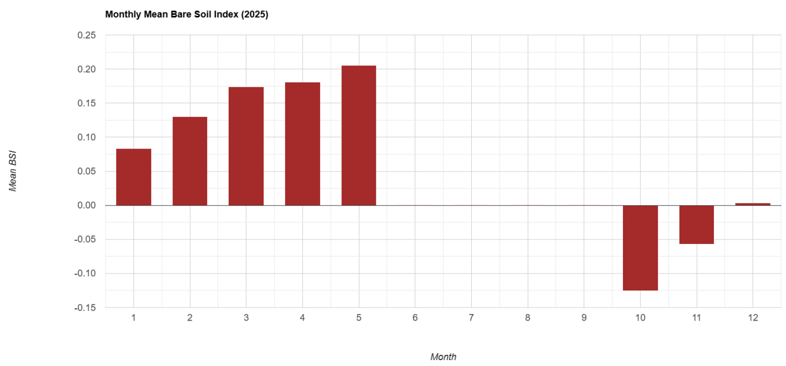


Figure 5.55: Monthly Mean BSI Trend for the Narmada River (Omkareshwar Stretch) for the year 2025.

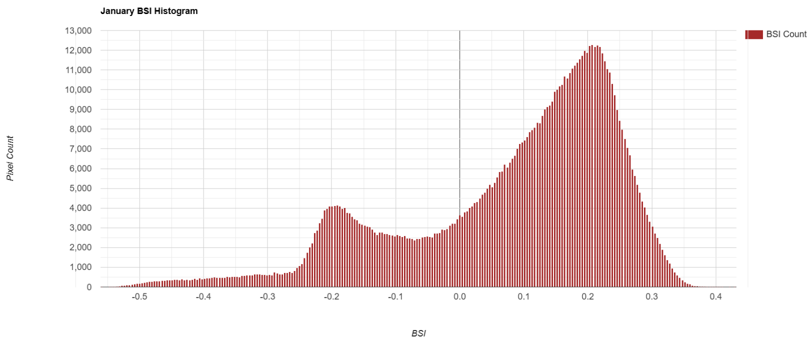
As illustrated in Figure 5.55, the mean BSI values trace a clear, ascending trajectory during the first half of the year. Starting at approximately 0.08 in January, the index rises consistently month over month, reaching a peak of roughly 0.20 in May. This upward trend perfectly captures the progressive drying of the basin, where receding water levels and dying vegetation increasingly expose bare soil, rocks, and sandbars.

Following the unrecorded monsoon period, October registers a drastically negative mean BSI (around -0.12). This signifies that the previously exposed soil is now heavily inundated by post-monsoon floodwaters or obscured by dense post-monsoon vegetation. A gradual recovery towards the baseline zero mark is observed through November and December as the floodwaters slowly recede.

5.7.5 Monthly Histogram and Statistical Analysis

To map the distribution of exposed soil pixels against water and vegetation, monthly histograms were computed. The following sections detail the pixel frequency distributions paired with their corresponding statistical summaries exported directly from GEE.

January 2025

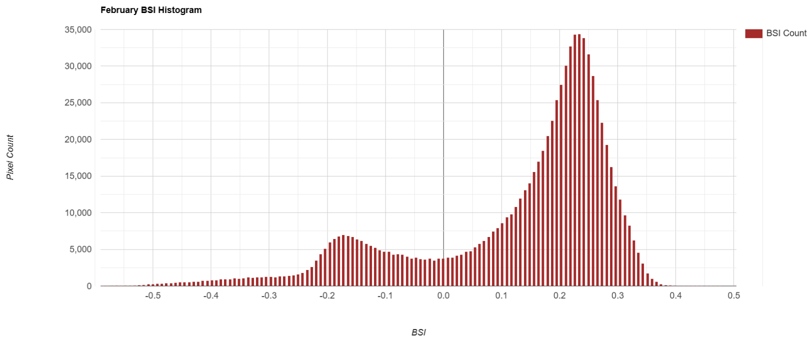


	A	B
1	Band Valu	BSI Count
2	-0.559	1
3	-0.555	
4	-0.551	
5	-0.547	3
6	-0.543	4
7	-0.539	7.31
8	-0.535	22
9	-0.531	34.671
10	-0.527	76
11	-0.523	75
12	-0.52	91.788
13	-0.516	107.718
14	-0.512	119.051
15	-0.508	151.141
16	-0.504	183.196
17	-0.5	182.396
18	-0.496	193.773

Figure 5.56: BSI pixel distribution histogram and statistical summary for January 2025.

The January histogram displays a distinct bimodal distribution. The smaller peak in the negative range represents the water channel, while the dominant, broader peak in the positive domain (around 0.2) indicates substantial exposure of dry, non-vegetated riverbanks and winter-exposed sandbars.

February 2025

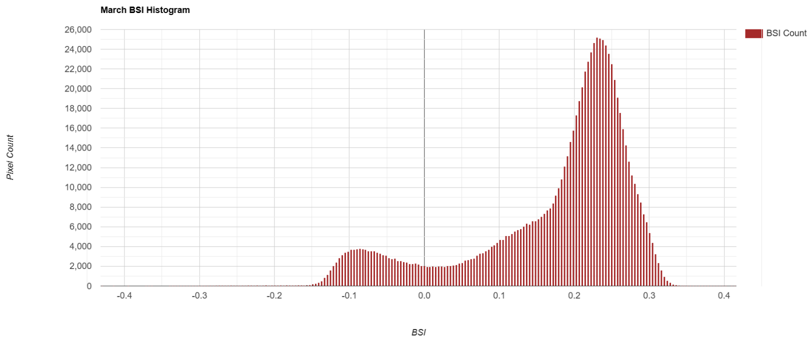


	A	B
1	Band Valu	BSI Count
2	-0.586	5
3	-0.578	24
4	-0.57	43
5	-0.563	50
6	-0.555	38
7	-0.547	56
8	-0.539	53.929
9	-0.531	70.576
10	-0.523	115.518
11	-0.516	165.973
12	-0.508	250.004
13	-0.5	250.859
14	-0.492	302.314
15	-0.484	351.455
16	-0.477	425.318
17	-0.469	436.145
18	-0.461	469.776

Figure 5.57: BSI pixel distribution histogram and statistical summary for February 2025.

In February, the histogram's positive peak intensifies and shifts slightly further to the right. The statistical summary reflects an increased mean BSI, signaling the early onset of summer drying, which leads to a higher proportion of exposed dry soil within the study area.

March 2025

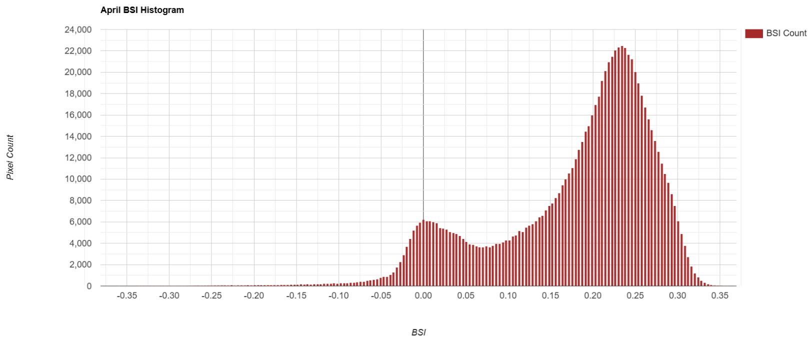


	A	B
1	Band Valu	BSI Count
2	-0.43	1
3	-0.426	
4	-0.422	
5	-0.418	
6	-0.414	1
7	-0.41	
8	-0.406	
9	-0.402	
10	-0.398	
11	-0.395	1
12	-0.391	
13	-0.387	
14	-0.383	
15	-0.379	
16	-0.375	
17	-0.371	1
18	-0.367	1

Figure 5.58: BSI pixel distribution histogram and statistical summary for March 2025.

The March data exhibits a continued rightward progression of the primary peak, moving beyond 0.25. As vegetative senescence accelerates (as noted in the NDVI analysis) and water levels drop, the spectral dominance of bare, rocky riverbeds and desiccated soil becomes highly apparent.

April 2025

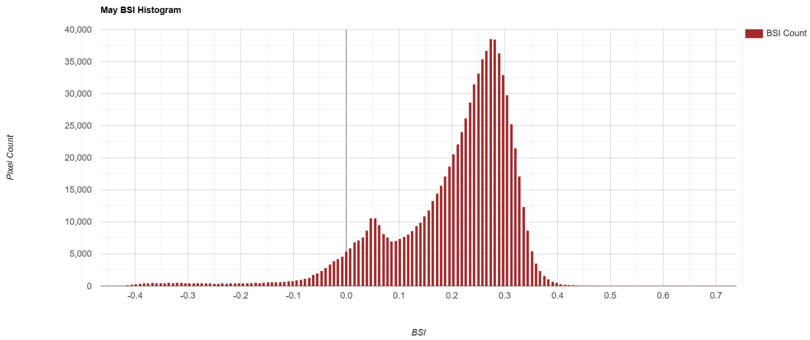


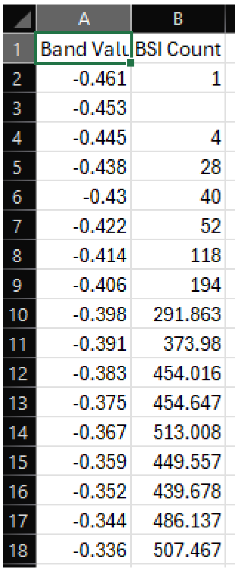
	A	B
1	Band Valu	BSI Count
2	-0.379	1
3	-0.375	
4	-0.371	
5	-0.367	
6	-0.363	
7	-0.359	1
8	-0.355	
9	-0.352	
10	-0.348	
11	-0.344	
12	-0.34	
13	-0.336	1
14	-0.332	2
15	-0.328	2
16	-0.324	2
17	-0.32	3
18	-0.316	15

Figure 5.59: BSI pixel distribution histogram and statistical summary for April 2025.

During April, the histogram showcases a massive concentration of pixels in the high positive range, while the negative water peak diminishes significantly in relative frequency. This denotes extensive sediment exposure and a severe contraction of the river's main channel.

May 2025





	A	B
1	Band Valu	BSI Count
2	-0.461	1
3	-0.453	
4	-0.445	4
5	-0.438	28
6	-0.43	40
7	-0.422	52
8	-0.414	118
9	-0.406	194
10	-0.398	291.863
11	-0.391	373.98
12	-0.383	454.016
13	-0.375	454.647
14	-0.367	513.008
15	-0.359	449.557
16	-0.352	439.678
17	-0.344	486.137
18	-0.336	507.467

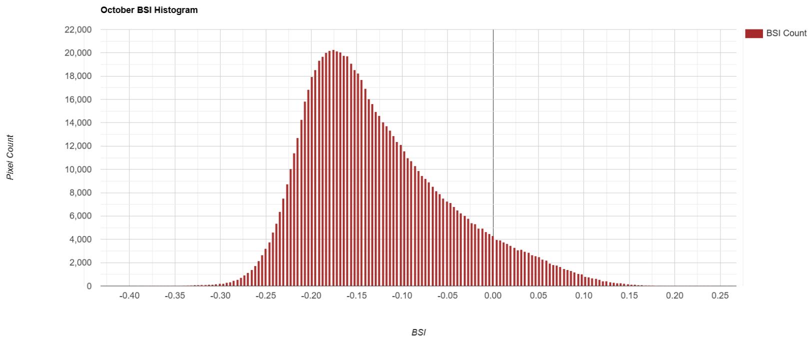
Figure 5.60: BSI pixel distribution histogram and statistical summary for May 2025.

May records the absolute peak of the Bare Soil Index. The histogram is overwhelmingly dominated by positive values, with the main peak shifting toward 0.3. This represents the maximum terrestrial exposure of the year, characteristic of peak pre-monsoon drought conditions in the Narmada Basin.

5.7.6 Monsoon Data Exclusions (June – September)

The temporal dataset intentionally excludes the months of June, July, August, and September. The calculation of the Bare Soil Index requires clean, interference-free optical measurements in the visible, NIR, and SWIR spectrums. The Indian southwest monsoon generates dense, continuous cloud cover that optical satellite sensors like Sentinel-2 cannot penetrate. Attempting to compute BSI during these months results in statistically invalid outputs driven by cloud reflectance rather than actual surface properties. Therefore, to uphold the rigorous scientific standards of this report, the monsoon quarter has been omitted.

October 2025

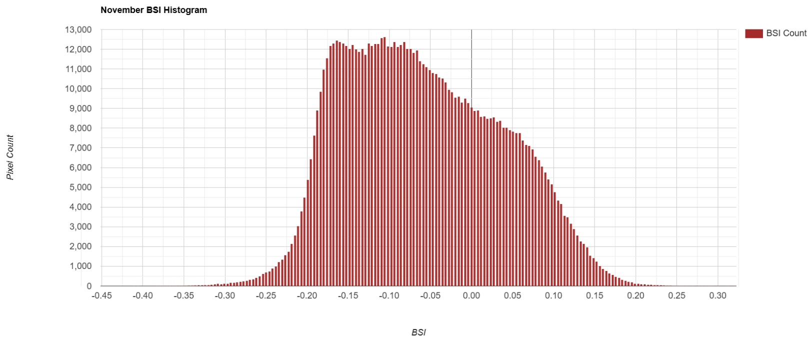


	A	B
1	Band Valu	BSI Count
2	-0.43	1
3	-0.426	
4	-0.422	
5	-0.418	
6	-0.414	
7	-0.41	
8	-0.406	
9	-0.402	
10	-0.398	
11	-0.395	
12	-0.391	1
13	-0.387	
14	-0.383	1
15	-0.379	1
16	-0.375	4
17	-0.371	1
18	-0.367	9

Figure 5.61: BSI pixel distribution histogram and statistical summary for October 2025.

Following the monsoon, the October histogram demonstrates a severe structural inversion. The distribution shifts almost entirely into the negative domain. This massive drop in BSI indicates that previously exposed bare soil is now completely covered by a combination of expanded river floodwaters and dense post-monsoon vegetation.

November 2025

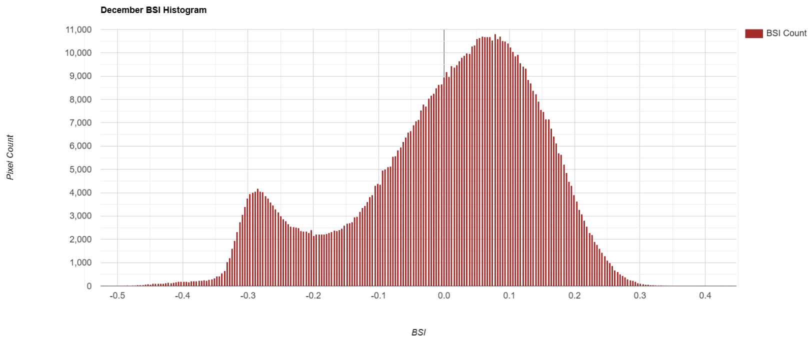


	A	B
1	Band Valu	BSI Count
2	-0.449	1
3	-0.445	
4	-0.441	
5	-0.438	
6	-0.434	
7	-0.43	
8	-0.426	
9	-0.422	
10	-0.418	
11	-0.414	
12	-0.41	1
13	-0.406	1
14	-0.402	
15	-0.398	
16	-0.395	
17	-0.391	
18	-0.387	

Figure 5.62: BSI pixel distribution histogram and statistical summary for November 2025.

In November, the distribution begins to flatten and spread back toward the zero baseline. As post-monsoon floodwaters begin their gradual recession and the surrounding soils lose their peak saturation, the spectral signature of bare land slowly begins to re-emerge along the ghat edges.

December 2025



	A	B
1	Band Valu	BSI Count
2	-0.523	1
3	-0.52	
4	-0.516	
5	-0.512	2
6	-0.508	
7	-0.504	4
8	-0.5	1
9	-0.496	2
10	-0.492	9
11	-0.488	12
12	-0.484	24
13	-0.48	10.722
14	-0.477	22.278
15	-0.473	23
16	-0.469	37.957
17	-0.465	33.224
18	-0.461	46

Figure 5.63: BSI pixel distribution histogram and statistical summary for December 2025.

The December analysis marks the closing of the hydrological cycle. The histogram exhibits the reformation of the classic bimodal distribution, with the mean BSI crossing back into positive territory. This confirms the stabilization of winter water levels and the renewed exposure of standard seasonal sandbars.

5.7.7 Discussion and Ecological Implications

The 12-month temporal tracking of the Bare Soil Index (BSI) offers a precise, quantitative measure of the morphological shifts within the Narmada River basin. The continuous rise in BSI from January to May acts as a direct metric for riverbed desiccation and habitat loss for shallow-water species.

Ecologically, highly elevated BSI values during the peak summer months indicate severe vulnerability to riverbank erosion. When the monsoon arrives, the dry, exposed, and non-vegetated topsoil mapped in May is highly susceptible to being washed away by high-velocity surface runoff, leading to increased sedimentation and turbidity (as corroborated by the NDTI findings). By mapping these high BSI zones prior to the monsoon, environmental engineers can pinpoint exact locations along Omkareshwar and Maheshwar that require artificial stabilization, thereby providing a data-driven approach to sustainable riverbank governance.

5.8 Total Suspended Matter (TSM)

5.8.1 Definition and Scientific Significance

Total Suspended Matter (TSM) refers to the organic and inorganic solid materials suspended in the water column, including silt, clay, plankton, and fine organic debris. In river monitoring, TSM is a primary indicator of water quality, erosion, and sedimentation rates. Unlike the Normalized Difference Turbidity Index (NDTI), which provides a relative qualitative measure of turbidity, TSM quantitative models attempt to estimate the actual physical concentration of suspended particles. In the Narmada Basin, tracking TSM is vital for assessing the impact of upstream dam discharges, soil erosion along the Omkareshwar and Maheshwar ghats, and the overall attenuation of light critical for benthic aquatic ecosystems.

5.8.2 Mathematical Formulation

TSM is typically derived using empirical or semi-analytical models calibrated to specific water bodies. For Sentinel-2 Level-2A imagery, the Red band (Band 4) exhibits a strong correlation with suspended particle concentration due to high scattering by suspended solids in this wavelength. A standard semi-analytical algorithm (such as the Nechad model) applied in remote sensing of water quality takes the form:

$$TSM = \frac{A \times Band\ 4\ (Red)}{1 - \frac{Band\ 4\ (Red)}{C}} + B \tag{5.8}$$

Where A, B, and C are calibration coefficients specific to the sensor's radiometric response and regional hydrodynamic properties. The output represents the concentration of suspended matter, reflecting the density of particulate scattering in the water column.

5.8.3 Methodology and Application

The TSM concentrations were mapped using the Google Earth Engine (GEE) platform. To ensure analytical accuracy, a stringent binary water mask (derived from MNDWI) was applied to isolate the river channel and eliminate terrestrial reflectance. Monthly median composites were generated to reduce localized atmospheric anomalies and sunglint. The algorithm was then applied pixel-by-pixel to evaluate the spatial gradient of suspended solids, identifying localized sedimentation zones and highly turbid flow channels.

5.8.4 Temporal Analysis and Observations

Tracking the mean TSM over time provides a direct visualization of the sediment loading cycle within the monitored stretches of the Narmada River.

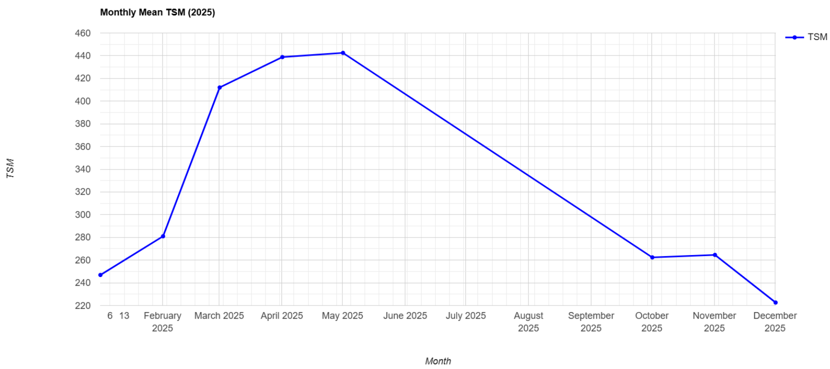


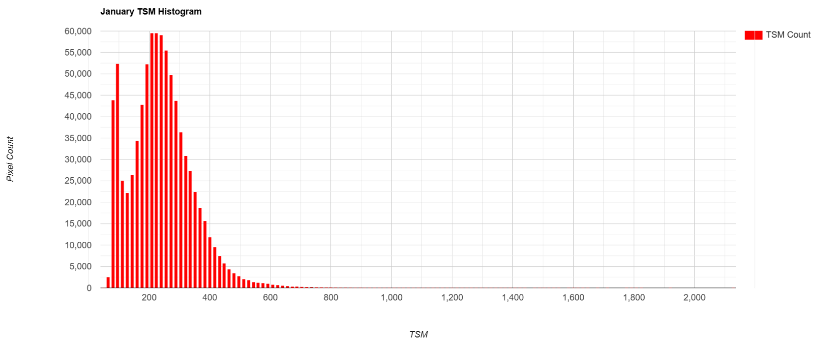
Figure 5.64: Monthly Mean TSM Trend for the Narmada River (Omkareshwar Stretch) for the year 2025.

The temporal trend illustrated in Figure 5.64 depicts a significant escalation in suspended matter during the pre-monsoon phase. The mean TSM value starts at a baseline low in January and February, reflecting calm, undisturbed water where sediments have settled. However, a sharp, almost exponential rise occurs through March, April, and May. This peak correlates with the lowest water volumes of the year, where even minimal hydrodynamic flow or localized anthropogenic activities along the ghats agitate the concentrated basal sediments, drastically driving up the suspended matter concentrations.

5.8.5 Monthly Histogram and Statistical Analysis

To map the distribution of suspended matter concentrations across the Omkareshwar and Maheshwar stretches, monthly pixel frequency histograms were generated. The distributions are paired with their respective statistical data exported from GEE.

January 2025

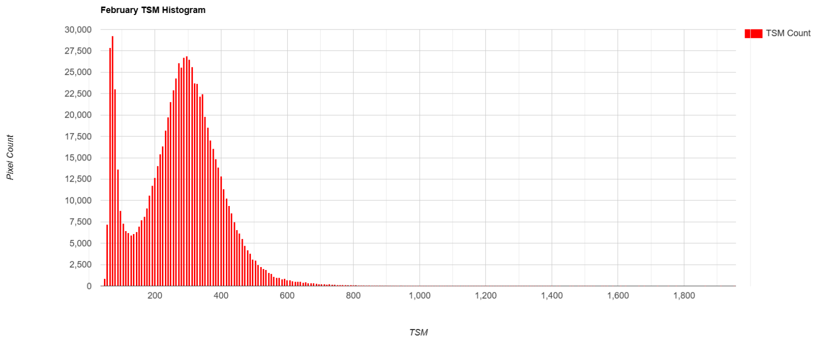


	A	B
1	Band Valu	TSM Count
2	48	9
3	64	2,512.09
4	80	43,921.09
5	96	52,364.01
6	112	25,142.40
7	128	22,188.20
8	144	26,542.68
9	160	34,420.80
10	176	42,850.30
11	192	52,239.57
12	208	59,528.16
13	224	59,503.19
14	240	59,061.19
15	256	55,529.97
16	272	49,740.74
17	288	43,740.38
18	304	36,422.97

Figure 5.65: TSM pixel distribution histogram and statistical summary for January 2025.

The January histogram displays a tight, narrow distribution on the lower end of the TSM spectrum. This confirms stable winter conditions characterized by low velocity and minimal turbulence, allowing suspended particles to deposit at the riverbed, resulting in exceptionally clear water.

February 2025

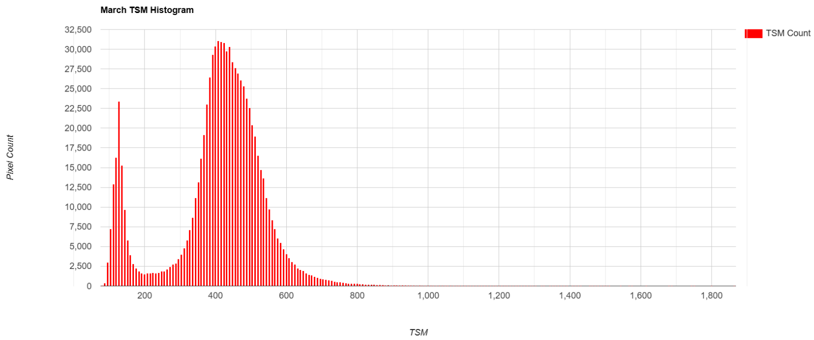


	A	B
1	Band Valu	TSM Count
2	40	17
3	48	860.886
4	56	7,176.44
5	64	27,844.16
6	72	29,274.43
7	80	23,044.77
8	88	13,648.21
9	96	8,789.45
10	104	7,314.00
11	112	6,424.15
12	120	6,244.71
13	128	5,944.39
14	136	6,094.51
15	144	6,327.20
16	152	6,953.18
17	160	7,714.90
18	168	8,110.95

Figure 5.66: TSM pixel distribution histogram and statistical summary for February 2025.

In February, the distribution remains concentrated at the lower threshold. The statistical summary records minimal mean TSM, aligning with the lowest turbidity observations (NDTI) of the year. The water body is relatively homogeneous in terms of clarity.

March 2025

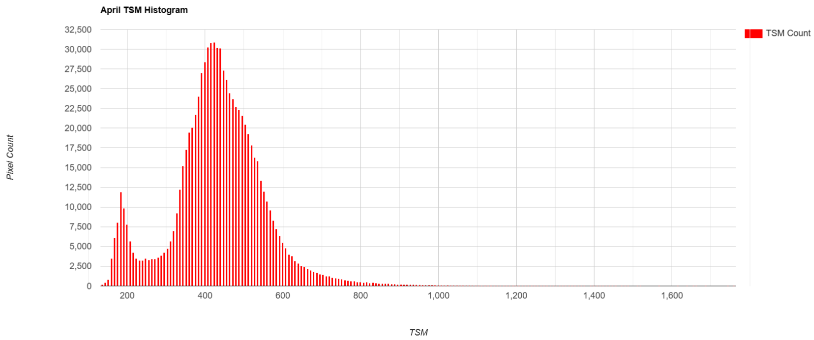


	A	B
1	Band Valu	TSM Count
2	80	41
3	88	389
4	96	2,985
5	104	7,208.48
6	112	12,932.28
7	120	16,279.82
8	128	23,377.74
9	136	15,275.78
10	144	9,683.86
11	152	5,798.44
12	160	3,950.78
13	168	2,818.79
14	176	2,217
15	184	1,854.24
16	192	1,638.70
17	200	1,500.37
18	208	1,638.75

Figure 5.67: TSM pixel distribution histogram and statistical summary for March 2025.

The March data illustrates a significant rightward shift and broadening of the histogram. As the dry season progresses and water volume decreases, the concentration of suspended matter begins to rise, leading to greater spatial variance in TSM values across the river channel.

April 2025

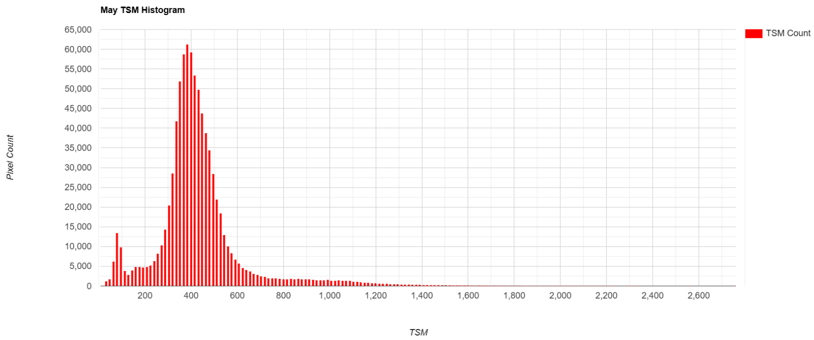


	A	B
1	Band Valu	TSM Count
2	136	185
3	144	448.906
4	152	803.478
5	160	3,514.17
6	168	6,107.48
7	176	8,022.64
8	184	11,901.64
9	192	9,847.28
10	200	7,767.46
11	208	5,706.03
12	216	4,247.17
13	224	3,507.91
14	232	3,267.05
15	240	3,241.64
16	248	3,467.78
17	256	3,294.19
18	264	3,420.78

Figure 5.68: TSM pixel distribution histogram and statistical summary for April 2025.

During April, the histogram shifts further into the higher concentration range. The CSV data confirms a rapid increase in the mean TSM. The reduced water column makes the river highly susceptible to bottom sediment resuspension from flow dynamics and wind-driven wave action.

May 2025



	A	B
1	Band Valu	TSM Count
2	16	8
3	32	1,240.02
4	48	1,768.67
5	64	6,258.70
6	80	13,460.04
7	96	9,898.29
8	112	3,889.18
9	128	2,881.08
10	144	4,004.46
11	160	4,908.66
12	176	4,925.64
13	192	4,711.95
14	208	4,926.75
15	224	5,234.58
16	240	6,353.64
17	256	8,179.64
18	272	10,329.54

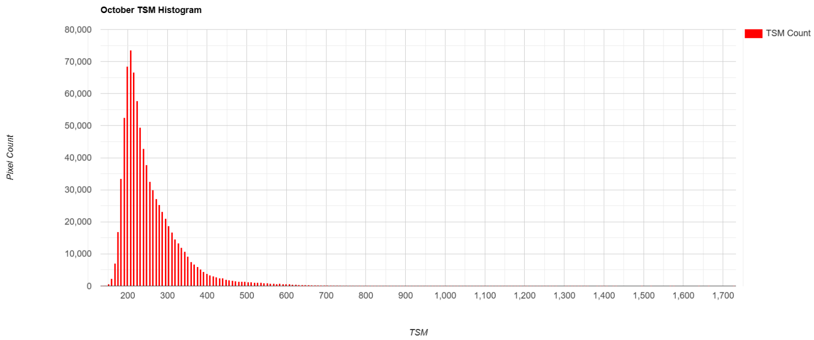
Figure 5.69: TSM pixel distribution histogram and statistical summary for May 2025.

May records the absolute peak of TSM prior to the monsoon. The histogram is heavily distributed toward the maximum values observed in the study. The high concentration of suspended matter confirms a highly turbid, sediment-laden aquatic environment characteristic of peak summer contraction.

5.8.6 Monsoon Data Exclusions (June – September)

The data for June, July, August, and September have been systematically excluded. The Total Suspended Matter algorithm relies heavily on precise optical reflectance from the Red band. During the Indian southwest monsoon, the Narmada Basin experiences dense and persistent cloud cover. Consequently, optical penetration by the Sentinel-2 sensor is heavily obstructed, introducing severe atmospheric scattering and rendering the TSM calculations statistically invalid. To preserve the scientific rigor of the report, this four-month block has been omitted.

October 2025

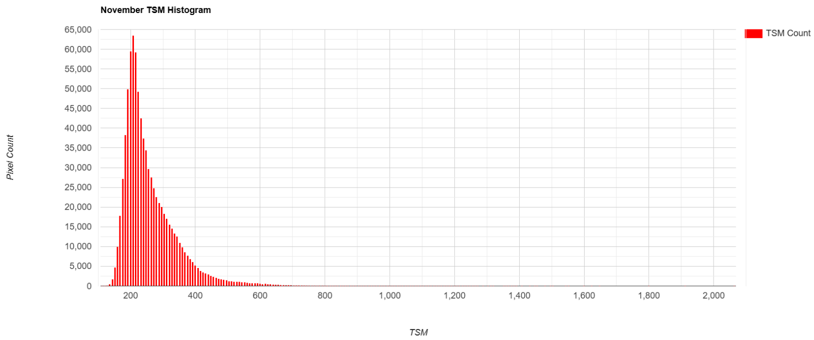


	A	B
1	Band Valu	TSM Count
2	136	1
3	144	90
4	152	581.235
5	160	2,369.51
6	168	6,997.05
7	176	16,924.51
8	184	33,407.09
9	192	52,476.40
10	200	68,487.23
11	208	73,558.59
12	216	66,584.37
13	224	57,743.40
14	232	49,473.06
15	240	42,914.50
16	248	37,756.35
17	256	32,485.15
18	264	29,952.27

Figure 5.70: TSM pixel distribution histogram and statistical summary for October 2025.

Following the monsoon, the October histogram reveals a broad distribution with elevated TSM levels. Despite the massive increase in river volume, the heavy inflow of terrestrial runoff, soil erosion, and suspended silt from the catchment area maintains a high sediment load, as confirmed by the elevated mean in the statistical export.

November 2025

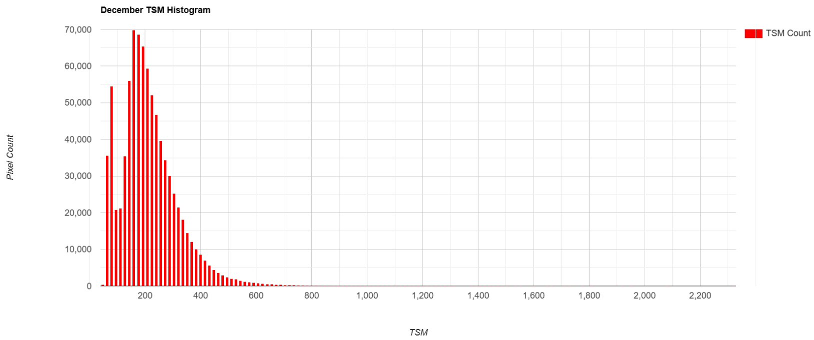


	A	B
1	Band Valu	TSM Count
2	112	2
3	120	19
4	128	99
5	136	525.267
6	144	1,691
7	152	4,769.40
8	160	9,967.82
9	168	17,864.52
10	176	27,171.48
11	184	38,315.69
12	192	49,905.71
13	200	59,559.91
14	208	63,505.66
15	216	59,310.01
16	224	49,335.78
17	232	42,530.58
18	240	37,483.47

Figure 5.71: TSM pixel distribution histogram and statistical summary for November 2025.

In November, the histogram demonstrates a sharp retraction toward the lower values. As the post-monsoon floodwaters slow down, their sediment-carrying capacity drops, allowing the heavier suspended particles to precipitate out of the water column, thereby gradually restoring water clarity.

December 2025



	A	B
1	Band Valu	TSM Count
2	48	380.012
3	64	35,547.00
4	80	54,523.85
5	96	20,766.89
6	112	21,175.69
7	128	35,472.49
8	144	56,054.55
9	160	69,859.80
10	176	68,698.75
11	192	65,372.18
12	208	59,424.41
13	224	52,123.51
14	240	46,689.71
15	256	39,682.75
16	272	34,459.20
17	288	30,084.36
18	304	25,311.83

Figure 5.72: TSM pixel distribution histogram and statistical summary for December 2025.

The December data marks the completion of the sedimentation cycle. The histogram fully reverts to a tight, narrow peak at the lower extreme of the axis, mimicking the January baseline. The CSV summary confirms stable, low-TSM conditions, indicating that the river has successfully flushed or deposited its monsoon sediment load.

5.8.7 Discussion and Ecological Implications

The 8-month tracking of Total Suspended Matter (TSM) highlights the extreme hydrodynamic and morphological stresses acting on the Narmada River.

From an ecological perspective, high TSM levels have profound implications for riverine health. During the peak summer (May) and immediately post-monsoon (October), the dense suspension of organic and inorganic matter dramatically increases light attenuation. This restriction of sunlight limits the photosynthetic capacity of benthic algae and submerged macrophytes, disrupting the primary food web. Furthermore, prolonged exposure to high suspended sediment concentrations can clog the gills of local fish species and smother spawning beds with fine silt as the water velocity drops. By quantitatively monitoring TSM, environmental authorities can make informed decisions regarding sediment flushing through dam gates and identify high-priority erosion zones requiring immediate stabilization along the ghats.